# REPORT 1 -
# NOSE CONE ANALYSIS

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**Weisshorn – Nose Cone Design Report**

EPFL ROCKET TEAM

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# Summary


| **Proposed Weisshorn nose cone** | |
|---|---|
| Profile | Von Kármán (Haack $C = 0$), tangent to the body |
| Length | $L = 810$ mm ($L/D = 4.5$) to the theoretical apex; 803.3 mm physical |
| Diameter | 180 mm |
| Tip | spherical, $r_n = 4.5$ mm ($r_n/R = 0.05$), machined Al 6082-T6 insert |
| Shell | CFRP $[0/45/0]_{SO}$, 1.2 mm (GFRP if the nose is the RF window), ≈354 g with tip |
| Fallback | same profile at $L/D = 4$ (720 mm) if the 62 mm cannot be found in the 5.4 m budget |

The Von Kármán profile is the member of the Haack family of noses that, according to slender-body theory, gives the lowest supersonic pressure drag for a given length and base diameter (Section 2.2). $L/D$ is the *fineness ratio*, the nose length divided by the body diameter. Symbols and abbreviations are listed in the Nomenclature.

Why:

- **Drag.** Above Mach 1 a shock wave forms ahead of the nose, and the overpressure it creates on the nose surface produces a pressure drag called *wave drag*. Wave drag is the only part of the vehicle drag that depends on the nose shape (Section 2.1). Seven profiles (cone, tangent ogive, parabola, power $\frac{1}{2}$ and $\frac{3}{4}$, LV-Haack, Von Kármán; Section 5) were compared with an axisymmetric Euler solver, i.e. a CFD code for inviscid compressible flow (validated to $< 0.5\,\%$ against Taylor–Maccoll, the exact solution for a cone, and grid-converged to 0.5 %). The solver was coupled to a trajectory model calibrated on the Launch Vehicle Overview (LVO). At $L/D = 4.5$, Von Kármán has the lowest wave drag at every computed Mach number. It flies 18 m to 22 m higher than the power noses, 69 m to 79 m higher than ogive, parabola and LV-Haack, and 132 m higher than the cone. It stays first in all seven robustness scenarios. At $-143\,\mathrm{m/kg}$ (apogee lost per kilogram of dry mass), the margin over the ogive is worth about 0.55 kg of dry mass.
- **Length.** $L/D = 4.5$ adds 60 m over the LVS length (the 720 mm nose of the Launch Vehicle Sizing) for 62 mm of nose bay, with diminishing returns beyond that: wave drag falls as $1/(L/D)^2$, while the friction and the mass of the nose grow linearly with its length.
- **Heating.** $T_0$, the stagnation temperature (the temperature air reaches when it is brought to rest against the tip, and the upper bound for any surface temperature), peaks at 119 °C (nominal) and 156 °C (hot-day envelope, $M = 1.56$). A printed PET-CF tip would exceed its heat-deflection temperature (the temperature at which the polymer starts to deform under load); the Al tip stays below 73 °C. The shell inner face reaches 104 °C in the envelope, close to the limit of the DP490 epoxy that bonds the coupler to the shell, so the thermal requirement must be raised from Firehorn's 390 K/5 s.
- **Integration and stability.** The tangent shoulder (no change of slope where the nose meets the body tube) suits the bonded coupler; 254 mm of the 3U payload fits inside the curved part. The static margin, the distance from the centre of gravity to the centre of pressure in body diameters (calibres, cal), changes by -0.04 cal, which is negligible.
- **Deliverables.** Profile coordinates, watertight STL solids of all seven designs and of the final

nose, the full Euler data set, ten numbered recommendations with owners (Section 7) and the alternatives to use if the priorities change.

**Not found in the ERT documentation** (details in Section 1.3): Alex Marseille's nose cone report, any Weisshorn nosecone requirement, the Weisshorn fin geometry and mass breakdown, and any supersonic flight data.

# Nomenclature

| Symbol | Meaning | Unit |
|---|---|---|
| *Geometry* | | |
| $D$, $R$ | body (base) diameter 180 mm, radius $R = D/2$ | m |
| $L$ | nose profile length, from the theoretical apex to the shoulder | m |
| $L/D$ | fineness ratio of the nose | – |
| $x$, $y$ | axial coordinate from the apex; local radius of the profile $y(x)$ | m |
| $\xi$, $\eta$ | normalised coordinates $\xi = x/L$, $\eta = y/R$ | – |
| $r_n$ | radius of the spherical tip (nose bluntness $r_n/R$) | m |
| $A_b$ | base area $\pi R^2$, also the reference area of all nose coefficients | $m^2$ |
| $V$ | internal volume of the nose (in Eq. (3): cell volume per radian) | $m^3$ |
| $\theta_{\mathrm{sh}}$ | angle between the profile and the body tube at the shoulder | ° |
| $n$, $K'$, $C$, $\theta$ | power-series exponent; parabolic-series parameter; Haack-series parameter and angle variable | – |
| *Flow* | | |
| $V_\infty$ | flight speed, i.e. the free-stream velocity seen from the vehicle | m/s |
| $M$, $M_\infty$ | Mach number $V/a$ (local flow speed over local speed of sound); free-stream Mach number $V_\infty/a_\infty$ | – |
| $M_{\max}$ | maximum Mach number of the flight | – |
| $a$ | speed of sound $\sqrt{\gamma R_g T}$ | m/s |
| $\gamma$, $R_g$ | ratio of specific heats (1.4 for air); specific gas constant of air (287 J/(kg K)) | –, J/(kg K) |
| $\mu$ | Mach angle, $\sin\mu = 1/M$ | ° |
| $p$, $p_\infty$ | static pressure; ambient (free-stream) pressure | Pa |
| $\rho$, $T$ | density (in Table 2 and Figure 13: radius of the ogive arc); static temperature | $kg/m^3$, K |
| $u$, $v$ | axial and radial velocity components | m/s |
| $E$ | total energy per unit volume, $p/(\gamma-1) + \frac{1}{2}\rho(u^2+v^2)$ | $J/m^3$ |
| $q_\infty$ | free-stream dynamic pressure $\frac{1}{2}\rho_\infty V_\infty^2 = \frac{1}{2}\gamma p_\infty M_\infty^2$ | Pa |
| $C_p$ | pressure coefficient $(p-p_\infty)/q_\infty$ | – |
| *Aerodynamic coefficients and flight* | | |
| $C_{D,\mathrm{w}}$ | nose wave-drag coefficient (reference area $A_b$) | – |
| $C_D$ | drag coefficient of the whole vehicle | – |
| $C_{D,f}$, $C_{D,\mathrm{fins}}$, $C_{D,\mathrm{base}}$, $C_{D,e}$ | skin-friction, fin pressure, base and excrescence drag coefficients | – |
| $k$ | calibration factor on the drag of everything except the nose | – |
| $C_N$, $C_{N\alpha}$ | normal-force coefficient; its slope with angle of attack $\mathrm{d}C_N/\mathrm{d}\alpha$ | –, $rad^{-1}$ |

| Symbol | Meaning | Unit |
|---|---|---|
| $\alpha$ | angle of attack (angle between the vehicle axis and the velocity) | rad |
| $x_{cp}$, $x_{cg}$ | position of the centre of pressure and of the centre of gravity | m |
| $F_{\mathrm{th}}$, $m$, $g_0$ | thrust; vehicle mass; standard gravity 9.81 m/s$^2$ | N, kg, m/s$^2$ |
| $I_t$, $I_{sp}$ | total impulse $\int F_{\mathrm{th}}\,\mathrm{d}t$; specific impulse $I_t/(m_p g_0)$ | N s, s |
| $m_p$ | propellant mass | kg |
| $z$, $\Delta z$ | altitude; apogee change relative to the baseline nose | m |
| *Heat transfer* | | |
| $T_0$ | stagnation (total) temperature | K |
| $T_r$, $r$ | recovery temperature; recovery factor | K, – |
| $T_\infty$, $T_w$ | ambient temperature; wall temperature | K |
| $h$, $h_s$ | convective heat-transfer coefficient; its value at the stagnation point | W/(m$^2$ K) |
| $c_p$ | specific heat at constant pressure | J/(kg K) |
| Pr, Re$_x$, St | Prandtl number $\mu_v c_p/k_t$; Reynolds number on the running length $x$; Stanton number $h/(\rho u c_p)$ | – |
| $\mu_v$, $k_t$ | dynamic viscosity; thermal conductivity | Pa s, W/(m K) |

| | |
|---|---|
| *Abbreviations* | |
| 3U | CubeSat payload unit, $100 \times 100 \times 340.5$ mm |
| cal | calibre: a length expressed in body diameters |
| CFD | computational fluid dynamics |
| CFRP, GFRP | carbon- and glass-fibre-reinforced polymer |
| CG, CP | centre of gravity, centre of pressure |
| CFL | Courant–Friedrichs–Lewy number (time step relative to the time a wave needs to cross a cell) |
| DJF, DDF | Design Justification File, Design Definition File (ERT documents) |
| DP490 | 3M Scotch-Weld DP490, the two-part epoxy used for the coupler bonds |
| EuRoC | European Rocketry Challenge |
| FD, ST, SE, AV, PL | ERT Flight Dynamics, Structure, Systems Engineering, Avionics and Payload teams |
| GCI | grid convergence index |
| HDT | heat-deflection temperature |
| HLLC | Harten–Lax–van Leer–Contact approximate Riemann solver |
| ISA | International Standard Atmosphere |
| LV, LVO, LVS | launch vehicle; Launch Vehicle Overview; Launch Vehicle Sizing |
| MUSCL | Monotonic Upstream-centred Scheme for Conservation Laws |
| PDR, CDR | Preliminary and Critical Design Review |
| PET-CF | carbon-fibre-filled PET, a 3D-printing filament |
| RANS | Reynolds-averaged Navier–Stokes equations |
| RF | radio frequency |
| SSP-RK2 | two-stage, strong-stability-preserving Runge–Kutta time integration |

# 1 Inputs, Requirements and Assumptions

## 1.1 What is fixed

Table 1 lists every input taken from the ERT documentation. The vehicle is **Weisshorn**, the 2027 competition launch vehicle flying in the EuRoC L9 category (target apogee 9 km). The sources are ERT documents: the Launch Vehicle Overview (LVO) and Launch Vehicle Sizing (LVS) of systems engineering, the requirement database (REQ), and the Design Justification and Definition Files (DJF, DDF) of earlier vehicles.

Table 1: Inputs from the ERT wiki, the Weisshorn requirements database and Firehorn heritage.

| Quantity | Value | Source |
|---|---|---|
| External diameter | 180 mm | task; 2027_C_SE_REQ_03 |
| Mission | EuRoC 2027, L9, 9 km apogee | Weisshorn wiki page; 2027_C_SE_LVO |
| LV length | $<$ 5.4 m (LVO: $\sim$ 5.4 m) | 2027_C_SE_REQ_04; 2027_C_SE_LVO |
| Dry / wet mass | $\sim$ 45 kg / $\sim$ 75 kg ($<$ 50 kg dry) | 2027_C_SE_LVO; 2027_C_SE_REQ_06 |
| Engine | EthaLOx, 4.5 kN, 9.1 s, $I_t = 40\,960$ N s, $I_{sp} = 200$ s | 2027_C_SE_LVO; 2027_C_SE_LVS_DJF |
| Maximum speed | Mach 1.4 | 2027_C_SE_LVO |
| Stability | 1.5–6 cal (EuRoC); FD target 2–4 cal | 2027_C_SE_REQ_09; 2027_C_SE_FD_REQ_07 |
| Launch rail | $84 \pm 1^\circ$ elevation | 2027_C_SE_FD_REQ_05 |
| Payload | 3U CubeSat in the nosecone bay, nose separates at apogee | 2027_C_SE_ST_FUA; Structure wiki page |
| Nose length (sizing) | scaled from Firehorn: $960 \times 180/240 = 720$ mm | 2027_C_SE_LVS_DJF (“scaled down in the same way for mass and length”) |
| RF windows | $360^\circ$, $\geq 2D$ long | 2027_C_SE_AV_INTERFACE_ST_REQ_07 |
| Manufacturing heritage | CFRP prepreg on PET-CF male mould, $[0/45/0]_{SO}$, 1.2 mm; removable PET-CF tip; bonded Al male coupler; as-built 500 g | 2024_C_ST_NOSECONE_DJF/DDF; 2025_C_ST_NOSECONE_PURPOSE_IR |
| Thermal heritage | 390 K for 5 s | 2024_C_SE_ST_NOSECONE_REQ_09 (Firehorn) |

The entries of Table 1 that drive the nose design are the following.

- **Maximum speed.** The Mach number $M = V_\infty/a$ compares the flight speed with the speed of sound, the speed at which small pressure disturbances travel through the air (about 332 m/s at 2 km in the ISA). With $M_{\max} \approx 1.4$, Weisshorn crosses the *transonic* range (roughly $0.8 < M < 1.2$, where subsonic and supersonic regions coexist around the body) and flies a few seconds in the low supersonic range. The nose has to be designed for this whole range, not for a single design Mach number.
- **Engine.** The total impulse $I_t$ is the thrust integrated over the burn; the specific impulse $I_{sp} = I_t/(m_p g_0)$ is the impulse delivered per unit weight of propellant. Together they fix the propellant

mass $m_p$, and with it the mass at burn-out that the nose design affects.

- **Stability.** A rocket is statically stable if its centre of pressure (CP, the point where the resultant aerodynamic normal force acts) lies behind its centre of gravity (CG): a small angle of attack then creates a moment that turns the nose back into the wind. The distance between the two, divided by the diameter, is the static margin in calibres (cal). The nose moves the CP forward, so its shape affects the margin (Section 2.3).
- **Payload.** A 3U CubeSat is a standard payload unit of $100 \times 100 \times 340.5$ mm. It sits in the nose bay, so the shape of the nose decides how much of it fits inside the curved part and how much extra cylinder (and vehicle length) is needed behind it.
- **RF windows.** Carbon fibre conducts electricity and blocks radio signals. Antennas for telemetry and GNSS therefore need a section of radio-transparent shell (glass fibre), and the nose is a candidate location for it.
- **Manufacturing heritage.** The Firehorn nose was laid up in prepreg (carbon fabric already impregnated with epoxy, cured in an oven) on the outside of a 3D-printed male mould. The laminate $[0/45/0]_{SO}$ is symmetric with an odd number of plies, i.e. five plies at $0°$, $45°$, $0°$, $45°$ and $0°$. The $0°$ plies, aligned with the axis, carry axial load and bending, and the $45°$ plies carry shear and torsion. The tip was a separate printed part, and an aluminium coupler is bonded inside the shoulder to connect the nose to the next bay.
- **Thermal heritage.** The Firehorn requirement asks the nose to withstand an outer surface temperature of 390 K for 5 s. Section 4.7 checks whether it covers the faster, lighter Weisshorn.

## 1.2 What is assumed

Everything below is an assumption of this study. Each one is chosen so that the comparison between nose shapes is fair; none of them changes the ranking.

- **Trajectory.** Point mass, rail 12 m at $84°$, ISA, no wind, constant 4.5 kN for 9.1 s. Propellant from $I_t/(I_{sp}g_0) = 20.9$ kg, i.e. burn-out mass 54.1 kg. The drag of everything except the nose is scaled by a single factor $k$ so that the baseline nose reaches the LVO apogee of 9.0 km (Section 3.3). This gives $M_{\max} = 1.47$, consistent with the LVO's Mach 1.4. A point-mass model treats the vehicle as a particle acted on by thrust, drag and gravity and ignores its rotation; for a statically stable rocket the axis stays aligned with the velocity, and this is the standard model for apogee estimates. The ISA (International Standard Atmosphere) gives temperature, pressure and density as functions of altitude. The factor $k$ is needed because the drag of the rest of the vehicle is the least certain part of the model at this stage (fins not designed, no flight data). Calibrating it once on the team's reference apogee means that every nose is compared on a vehicle with the intended performance. Since all noses share the same $k$, the ranking depends only on their differences, and Section 4.5 shows that it does not change when the rest of the drag is varied.
- **Thermal envelope.** The LVO masses ($75 \to 45$ kg) on a 30 °C day give $M_{\max} \approx 1.56$. Heating is checked on this case. A lighter vehicle reaches a higher speed, and hotter ambient air raises the stagnation temperature for a given Mach number (Section 3.4); together they give the hottest credible flight.

- **Fins** (not yet defined for Weisshorn): four trapezoidal fins, root 300 mm, tip 120 mm, span 160 mm, sweep 180 mm, 5 mm hexagonal section. They only enter the common drag and the CP shift; they do not change the ranking of the noses. Fins provide most of the restoring normal force of the vehicle and therefore set where its CP lies.
- **Payload envelope.** A 3U unit ($100 \times 100 \times 340.5$ mm) needs an outer radius $r \geq 77$ mm: 70.7 mm half-diagonal ($100/\sqrt{2}$, the radius of the circle through the corners of its square section), 3 mm clearance, 2 mm guide, 1.2 mm shell.
- **Shell.** 1.2 mm CFRP at the Firehorn as-built areal density (0.98 kg/m$^2$, the mass per unit of wetted surface), plus a 60 g machined aluminium tip.
- **Transonic nose wave drag** ($0.8 < M < 1.1$): smooth rise from zero at $M = 0.8$ to the computed $M = 1.1$ value at $M = 1.0$ (Section 3.2).

### 1.3 What could not be found

- **Alex Marseille's nose cone report** is not on the ERT Drive or the wiki. The searches covered "Marseille", "nose cone", "nosecone", "ogive", "SSR" and "Farewell". His documented work is the CFD of the 54 mm supersonic semester rocket (SSR weekly meetings 01–12, spring 2026; report on Overleaf, not shared in the Drive). As CFD advisor he set the semester goal "WS (Cd, Nosecone)" (2027_C_FD_TM_3, 01.10.2026), which is this study. The heritage used instead is the 2021 supersonic nosecone semester project (materials and tip, ERT Drive, *Report* folder), the Wildhorn optimisation note (2022_FD_TN_C01) and the Firehorn nosecone DJF/DDF.
- **No Weisshorn nosecone requirement** (length, mass, thermal) exists yet. The four Weisshorn nosecone pages (DJF, DDF, FEAR, tooling DJF) are empty templates. The nosecone requirements of the 2027 database cover only the Aletsch test rocket.
- **Weisshorn fins, CG and mass breakdown** are not defined (FD page is a template). The static margin is therefore given as a *change* relative to the baseline nose.
- **No supersonic flight data** exists at ERT. Firehorn 1 reached 223 m, so there is no measured nose heating or drag to calibrate against.

## 2 Design Options

The nose shape matters for Weisshorn because of the way air flows around a body above Mach 1. This section first summarises that physics (Section 2.1), then defines the candidate profiles (Section 2.2) and the criteria on which they are compared (Section 2.3).

### 2.1 Supersonic flow over a nose and wave drag

**Subsonic and supersonic flow.** In subsonic flow, the pressure disturbances created by the body travel upstream faster than the body moves. The oncoming air therefore starts to move aside well ahead of the nose and the streamlines adjust smoothly. As long as the flow stays attached, the overpressure on the front of a streamlined nose is largely recovered on its aft part, so its drag is mostly skin friction. Above $M = 1$, disturbances can no longer travel upstream:

a small disturbance created at a point only influences the region inside a downstream cone of half-angle

$$\mu = \arcsin \frac{1}{M}, \tag{1}$$

the *Mach angle* ($\mu = 45.6°$ at $M = 1.4$). The air ahead of the nose receives no advance signal and is deflected abruptly through a *shock wave*, a thin layer across which pressure, density and temperature jump and the velocity drops. Entropy increases across a shock: part of the kinetic energy of the flow is dissipated irreversibly. Where the surface curves back towards the axial direction, the flow turns through a continuous *expansion* (a Prandtl–Meyer expansion), in which the pressure decreases, possibly below the ambient value.

**Wave drag.** The pressure on a supersonic nose is therefore above ambient near the tip and decreases along the nose. The nose surface faces forward everywhere, so this overpressure gives a net axial force, the *wave drag*. It exists even in an inviscid fluid. Physically it is the energy carried away by the waves and dissipated in the shock. It is expressed as the coefficient $C_{D,\mathrm{w}} = D_\mathrm{w}/(q_\infty A_b)$, where $D_\mathrm{w}$ is the wave-drag force, $q_\infty = \frac{1}{2}\rho_\infty V_\infty^2$ the free-stream dynamic pressure and $A_b = \pi R^2$ the base area (Equation (4)). Wave drag appears once the local flow becomes sonic somewhere on the nose, around $M \approx 0.8$ for slender bodies. It rises steeply to a peak near $M = 1$ and decreases slowly in the supersonic range. Of the drag components of a rocket, it is the only one controlled by the nose shape: skin friction depends on the wetted area, and base drag on the aft end (Section 3.3).

**Attached and detached shocks.** A sharp, slender tip deflects the flow by a small angle, so the shock is weak, oblique and attached to the tip. If the tip is blunt, or its half-angle too large for the Mach number, no attached shock exists. The shock then detaches and stands ahead of the body as a curved *bow shock*. Its central part is a normal shock, behind which the flow is subsonic and the pressure rises almost to the stagnation pressure. This is why supersonic noses are pointed, and why any rounding of the tip has to stay small (Section 4.6).

**Slender-body theory.** For a slender nose ($L/D \gg 1$) the disturbances are small and the compressible potential-flow equation can be linearised. In this theory the wave drag depends only on the axial distribution of cross-sectional area $S(x) = \pi y(x)^2$, through its second derivative $S''(x)$, i.e. through how abruptly area is added along the nose. Two results follow. First, within a profile family all radii scale with $R$ and all lengths with $L$, so $S'' \propto R^2/L^2$ and

$$C_{D,\mathrm{w}} \propto \left(\frac{R}{L}\right)^2 \propto \frac{1}{(L/D)^2} : \tag{2}$$

doubling the length of a nose divides its wave drag by four. Second, the area distribution of minimum wave drag can be found analytically. For a given length and base area it is the Von Kármán profile, with $C_{D,\mathrm{w}} = 1/(L/D)^2$. For a given length and volume it is the LV-Haack profile (Section 2.2). The theory assumes small disturbances and replaces shocks by Mach waves, so it gives trends rather than exact values, and it is least accurate near $M = 1$. The Euler computations of Section 3 provide the values used in this study.

## 2.2 Candidate profiles

The nose is defined by three choices: the **profile family**, the **fineness** $L/D$ (curved length over base diameter) and the **tip bluntness** $r_n/R$. Seven profiles are compared (Table 2, Figure 2); Figure 1 sketches each of them and how it is constructed. With $\xi = x/L$ (the axial coordinate from the apex, normalised by the length) and $\eta = y/R$ (the local radius, normalised by the base radius), they are:

- **Conical**, $\eta = \xi$: a straight generator of half-angle $\arctan(R/L)$ ($7.1°$ at $L/D = 4$). It is the simplest shape, but it meets the body tube with a kink equal to its half-angle.
- **Tangent ogive**: the generator is a circular arc of radius $\rho = (R^2 + L^2)/(2R)$ whose centre lies in the base plane, so that the arc is tangent to the body tube at the shoulder. It is the traditional subsonic rocket nose and the shape of earlier ERT vehicles.
- **Parabolic series**, $\eta = (2\xi - K'\xi^2)/(2 - K')$; the case $K' = 1$ used here is the parabola that is tangent to the body at the shoulder.
- **Power series**, $\eta = \xi^n$ with $0 < n < 1$ ($n = 1$ is the cone). The smaller $n$, the fuller the front of the nose. For $n < 1$ the slope $\mathrm{d}\eta/\mathrm{d}\xi = n\xi^{n-1}$ is infinite at the apex. For $n = \frac{1}{2}$ the profile is a parabola about the axis, $y^2 = R^2 x/L$, whose apex has a radius of curvature $R^2/(2L)$ (5.6 mm at $L/D = 4$): this nose is blunt by construction. At the shoulder the slope is $nR/L \neq 0$, so the power noses meet the body with a kink $\theta_{\mathrm{sh}} = \arctan(nR/L)$.
- **Haack series**: the area distributions of minimum wave drag from slender-body theory. With the variable $\theta = \arccos(1 - 2\xi)$, which runs from 0 at the apex to $\pi$ at the base,

$$\eta = \frac{1}{\sqrt{\pi}} \sqrt{\theta - \tfrac{1}{2}\sin 2\theta + C \sin^3 \theta}.$$

  The parameter $C$ selects the constraint: $C = 0$ gives the minimum wave drag for a given length and base area (**Von Kármán**), $C = \frac{1}{3}$ the minimum for a given length and volume (**LV-Haack**). Both are tangent to the body at the shoulder. Near the apex the Von Kármán profile grows as $\eta \propto \xi^{3/4}$, with a vertical tangent but a zero tip radius.

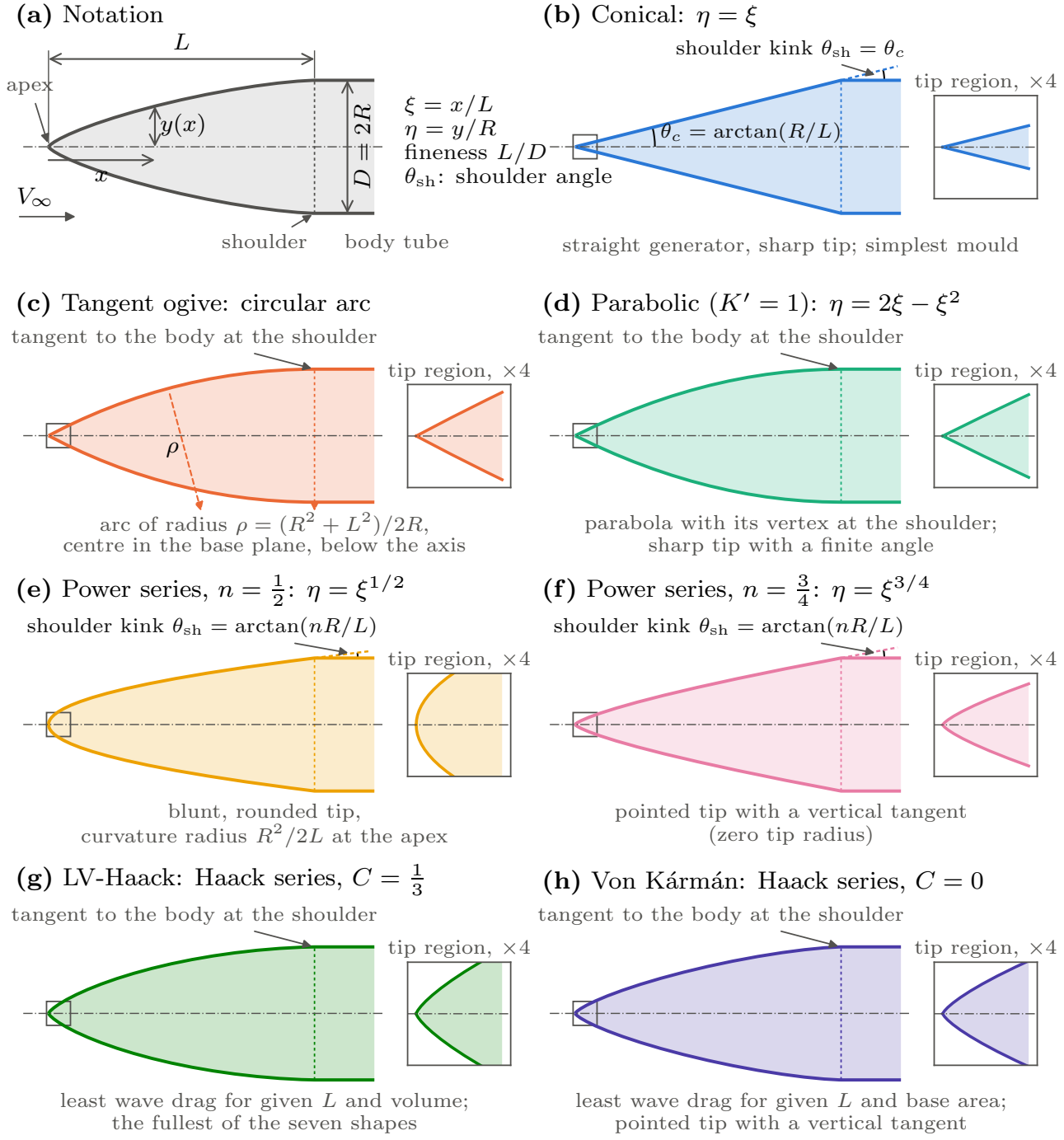



Figure 1: Schematics of the candidate profiles, drawn at $L/D = 2$ (half the design fineness) so that their differences are visible; Figure 2 compares them at real proportions. (a) Notation. (b)–(h) Construction of each profile: the upper label describes the shoulder (tangent to the body, or a kink of angle $\theta_{sh}$), the lower label the generator and the tip, and the inset shows the boxed tip region magnified four times.

Figure 1 shows the three features that distinguish the profiles. The first is how the nose meets the body tube: tangent for the ogive, parabola and Haack noses, with a kink for the cone and the power noses. The second is the shape of the tip: a finite-angle point (cone, ogive, parabola), a pointed tip with a vertical tangent (power $\frac{3}{4}$, Haack noses) or a rounded, blunt tip (power $\frac{1}{2}$). The third is how the area is distributed along the length, from the thin cone to the full LV-Haack nose.

Table 2: Candidate profiles. $x_{cp}$: Barrowman nose centre of pressure; $\theta_{\mathrm{sh}}$: angle to the body at the shoulder (0 = tangent).

| Profile | $\eta(\xi)$ | ERT heritage | $x_{cp}/L$ | $\theta_{\mathrm{sh}}$ at $L/D = 4$ |
|---|---|---|---|---|
| Conical | $\xi$ | – | 0.667 | $7.1^\circ$ |
| Tangent ogive | $\sqrt{\rho^2 - (L-x)^2} + R - \rho$, $\rho = \frac{R^2+L^2}{2R}$ | Bella Lui II, Nordend | 0.464 | $0^\circ$ |
| Parabolic ($K' = 1$) | $2\xi - \xi^2$ | – | 0.467 | $0^\circ$ |
| Power series $n = \frac{1}{2}$ | $\xi^{1/2}$ | Wildhorn, Firehorn | 0.500 | $3.6^\circ$ |
| Power series $n = \frac{3}{4}$ | $\xi^{3/4}$ | – | 0.600 | $5.4^\circ$ |
| LV-Haack ($C = \frac{1}{3}$) | $\pi^{-1/2}\sqrt{\theta - \frac{1}{2}\sin 2\theta + C\sin^3\theta}$, $\theta = \arccos(1-2\xi)$ | – | 0.438 | $0^\circ$ |
| Von Kármán ($C = 0$) | as LV-Haack with $C = 0$ | – | 0.500 | $0^\circ$ |

The tangent-ogive entry of Table 2 is written in dimensional form ($y$ in place of $\eta$, with $\rho$ the arc radius). The last two columns are used later: $x_{cp}/L$ in the stability assessment (Section 2.3) and $\theta_{\mathrm{sh}}$ in the manufacturing one.

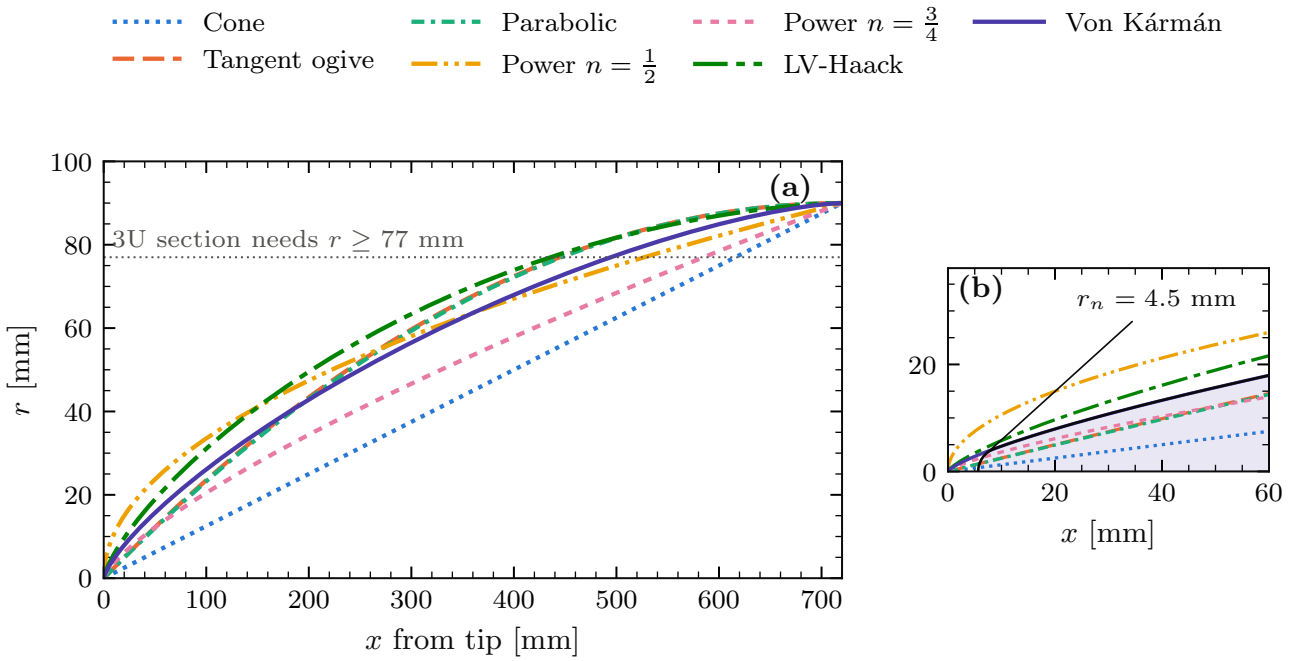



Figure 2: (a) The seven candidate profiles at $L/D = 4$ ($L = 720$ mm), with the radius a 3U section needs. (b) Tip region; the shaded profile is the Von Kármán nose with a 4.5 mm spherical tip ($r_n/R = 0.05$) selected in Section 6.

At equal length the profiles differ mainly in how quickly they gain radius (Figure 2a). The power-$\frac{1}{2}$ nose is the fullest over the first fifth of the length. The ogive, parabola and LV-Haack are the fullest over the middle and aft part, and the cone is the thinnest everywhere. Each curve's length above the dotted line $r = 77$ mm is the part of the nose that can house the 3U payload. In Figure 2b the apex of the Von Kármán nose is replaced by a 4.5 mm sphere; the change is confined to the first 10 mm of the nose.

Shapes discarded before computing: the *elliptical* nose (good only subsonically; its blunt front gives a strong bow shock above Mach 1) and the *secant ogive* (no advantage over the tangent ogive and LV-Haack at the same volume, and a slope discontinuity at the shoulder). The elliptical nose is a quarter-ellipse whose front is a rounded, blunt face; above Mach 1 it carries a detached bow shock with the near-stagnation pressure described in Section 2.1. The secant ogive is a circular arc of larger radius than the tangent ogive, cut so that it meets the body at an angle.

### 2.3 Comparison criteria

What matters for the comparison:

- **Wave drag.** At Weisshorn's Mach numbers (up to 1.5) the nose pressure drag depends mostly on $L/D$ and on how much area the profile adds near the tip. Slender-body theory gives the Von Kármán profile the minimum wave drag for given length and base area, $C_{D,\mathrm{w}} = 1/(L/D)^2$. This makes the Von Kármán nose the theoretical favourite. Whether this linear-theory result survives at Weisshorn's finite thickness and Mach numbers is what the Euler computations of Section 4 check.
- **Volume and payload.** Fuller profiles (ogive, parabolic, LV-Haack) reach $r = 77\,\mathrm{mm}$ further forward, so more of the 3U unit sits inside the curved part. The part of the 3U that does not fit inside the curved nose needs a cylindrical extension of the nose bay, which adds vehicle length within the 5.4 m cap.
- **Stability.** The nose carries $C_{N\alpha} = 2$ whatever its shape. Its centre of pressure sits at $x_{cp} = L - V/A_b$, so a fuller nose moves the vehicle CP forward. At a small angle of attack $\alpha$, each part of the vehicle produces a force normal to its axis, $N = q_\infty A_b C_{N\alpha}\,\alpha$. Slender-body theory gives the same slope $C_{N\alpha} = 2$ per radian for any nose that closes on the base area $A_b$, because the normal force generated by a section is proportional to its rate of change of area $\mathrm{d}S/\mathrm{d}x$, and its integral over the nose only depends on the base area. The shape only sets where the force acts, $x_{cp} = L - V/A_b$ from the apex ($2L/3$ for a cone). The vehicle CP is the $C_{N\alpha}$-weighted average of the nose and fin contributions, and the static margin is $(x_{cp} - x_{cg})/D$; moving the CP forward reduces it.
- **Manufacture.** A tangent shoulder ($\theta_{\mathrm{sh}} = 0$) avoids a kink at the coupler joint. A sharp tip cannot be laid up in prepreg (ply overlaps stack at the apex), so every option uses a separate machined tip. A kink would also create a local expansion in the flow, need a fillet in the mould, and put the bond line of the coupler at a corner of the shell.

## 3 Method

The chain is: Euler wave drag of each nose $\rightarrow$ vehicle drag build-up $\rightarrow$ ascent trajectory $\rightarrow$ apogee, heating, stability. All scripts are in `scripts/` (Appendix A). The wave drag is the only shape-dependent quantity, but a drag coefficient alone does not say how much a shape is worth. It is therefore converted into apogee with a flight model, which also accounts for the length and mass of each nose, and the same trajectories drive the heating assessment.

### 3.1 Nose wave drag: axisymmetric Euler solver

Above Mach 1 the nose pressure drag is an inviscid effect (shock and expansion), so the axisymmetric Euler equations are solved for each nose followed by a $5R$ cylinder. The Euler equations express the conservation of mass, momentum and energy of a compressible fluid without viscosity and heat conduction, i.e. the Navier–Stokes equations without their viscous terms. Neglecting viscosity is appropriate for the wave drag, which is set by the pressure field of the outer flow. The boundary layer is thin at these Reynolds numbers (about $2 \times 10^7$ on the nose length at $M_{\max}$), and its main effect, skin friction, is added separately in the flight model (Section 3.3). At zero

angle of attack the flow is *axisymmetric*: it is the same in every plane through the axis, so only the half-plane $(x, r)$ is computed. In finite-volume form, for one cell and per radian of azimuth,

$$\frac{\partial(\boldsymbol{U}V)}{\partial t} + \sum_{\text{faces}} \boldsymbol{F}\cdot\boldsymbol{n}\,\ell\, r_f = \begin{pmatrix} 0 & 0 & pA & 0 \end{pmatrix}^{\top}, \qquad V = A\, r_c, \tag{3}$$

with $\boldsymbol{U} = (\rho, \rho u, \rho v, E)$ per radian. $\boldsymbol{U}$ is the vector of conserved variables: density, axial and radial momentum per unit volume, and total energy per unit volume. $\boldsymbol{F}$ is the corresponding flux, the rate at which each quantity crosses a face, including the pressure force acting on it. $\boldsymbol{n}$ is the outward unit normal of a face, $\ell$ its length in the $(x, r)$ plane and $r_f$ the radius of its midpoint, so $\ell\, r_f$ is the face area per radian. $A$ is the area of the cell in the $(x, r)$ plane and $r_c$ the radius of its centroid, so $V = A r_c$ is the volume per radian of the ring-shaped cell. The equation states that the content of a cell changes by the net inflow through its faces plus a source. The source $p\,A$ appears in the radial momentum equation only. The two azimuthal faces of a ring-shaped cell are not parallel, so the pressure acting on them has a net outward radial component, which has no equivalent in planar 2D flow.

The scheme is a cell-centred finite-volume method with HLLC fluxes, MUSCL reconstruction of primitive variables (minmod), SSP-RK2 and local time stepping (CFL 0.85). Writing the volume as $Ar_c$ preserves a uniform flow exactly. In detail:

- *Finite volume*: the unknowns are cell averages of $\boldsymbol{U}$, updated with the fluxes through the faces. Mass, momentum and energy are conserved exactly, which places shocks at the right position.
- *HLLC flux*: at each face, the two neighbouring gas states form a local Riemann problem (a jump between two states), whose solution is a set of waves travelling at speeds related to $u - a$, $u$ and $u + a$. The HLLC solver approximates this pattern with three waves and returns the flux through the face. It resolves shocks within two to three cells without oscillations.
- *MUSCL with minmod*: inside each cell the variables are reconstructed as linear functions, which makes the scheme second-order accurate in space. The minmod limiter flattens these slopes near discontinuities, so that no spurious oscillations appear at shocks.
- *SSP-RK2 and local time stepping*: a two-stage Runge–Kutta integration that keeps this non-oscillatory property. Only the steady state matters, so each cell advances with its own largest stable time step. The CFL number is the ratio of that time step to the time a wave needs to cross the cell; an explicit scheme is stable only for values below one.

The grid is body-fitted ($160 \times 80$ cells), starts on the stagnation streamline upstream of the tip, keeps near-wall cells orthogonal to the body and clusters points at the tip and at the shoulder. The outer boundary is a hyperbola outside the bow shock with characteristic (Riemann-invariant) far-field conditions; the wall is a slip wall and the axis a symmetry line (Figure 3). A body-fitted grid has one family of grid lines along the wall, so the wall condition is applied exactly. The outer boundary must lie upstream of the bow shock so that the flow entering the domain is the undisturbed free stream. The characteristic conditions impose free-stream values on the waves entering the domain and let outgoing waves leave without reflection. In inviscid flow the only condition at a solid wall is that no fluid crosses it; the velocity is free to slip along it.

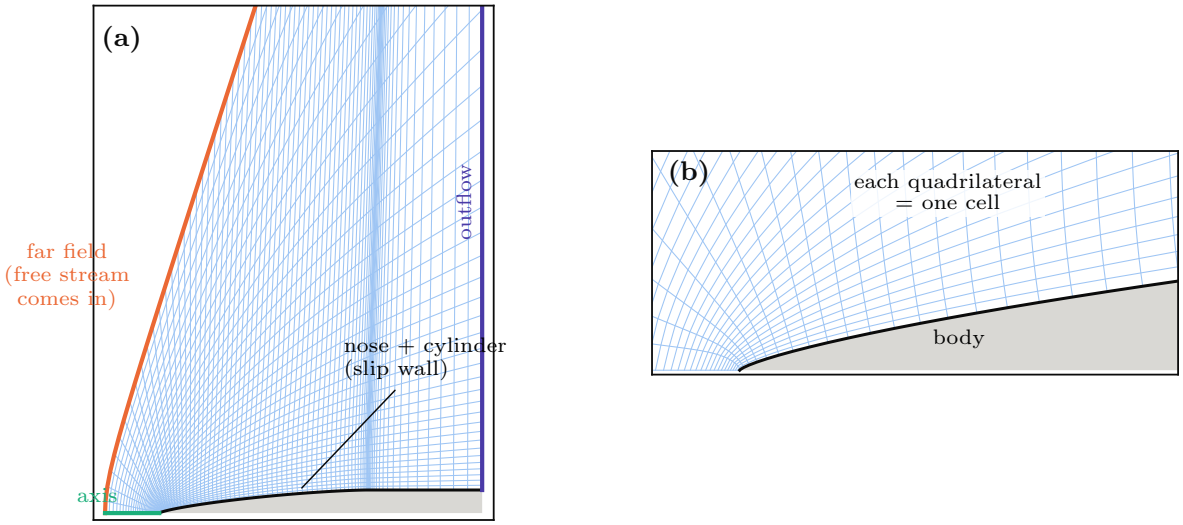



Figure 3: Computational domain and grid around a nose (coarse grid for clarity; the production grid has twice as many cells in each direction). (a) Boundaries: far field upstream of the bow shock, outflow plane, axis of symmetry, and slip wall along the nose and the cylinder. (b) Close-up of the tip region.

The drag coefficient is the integral of the wall pressure over the nose,

$$C_{D,\mathrm{w}} = \frac{1}{q_\infty \pi R^2} \int_{\mathrm{nose}} (p - p_\infty)\,(-n_x)\,2\pi r\,\mathrm{d}s, \tag{4}$$

and a run is converged when $C_{D,\mathrm{w}}$ changes by less than $2 \times 10^{-5}$ (relative) over 250 iterations. In Equation (4), $p - p_\infty$ is the wall overpressure, $n_x$ the axial component of the outward wall normal (negative on a forward-facing surface, so $-n_x > 0$), $2\pi r\,\mathrm{d}s$ the area of a surface ring of width $\mathrm{d}s$ at radius $r$, and $q_\infty \pi R^2$ the reference force. Only the pressure difference matters, since a uniform pressure gives no net force on a closed body. The convergence threshold is far below the differences between shapes. The Mach numbers are computed from high to low (1.6, 1.4, 1.2, 1.1; also 1.3 for the cone and ogive), each started from the previous solution. Starting from a converged neighbouring solution shortens the transient, because the shock system only has to move, not form. Convergence slows as the Mach number decreases: the bow shock stands further ahead and the subsonic region behind it grows, and in a subsonic region disturbances propagate in all directions instead of being swept out of the domain. The $M = 1.1$ runs, with a large subsonic region behind the bow shock, were stopped at 6000 iterations before meeting this criterion. They only set the end point of the transonic model (Section 3.2) and carry a larger uncertainty, estimated at 1–2 %. Parabolic and power-$\frac{3}{4}$ were computed at $L/D = 4$ only and scaled with $1/(L/D)^2$; LV-Haack and the ogive at $L/D = 5$ were stopped at $M = 1.2$.

**Validation.** For a sharp cone the surface pressure is uniform and $C_{D,\mathrm{w}} = C_p$, which is known exactly from the Taylor–Maccoll equation. For a sharp cone at zero incidence the shock is a straight conical surface attached to the apex, and the flow between shock and cone is *conical*: every property is constant along rays from the apex. The Euler equations then reduce to an ordinary differential equation in the ray angle, the Taylor–Maccoll equation, which is integrated numerically to high accuracy (`scripts/taylor_maccoll.py`). Because the overpressure is uniform, Equation (4) reduces to $C_{D,\mathrm{w}} = C_p$. Figure 4a compares the two for four cone angles (7.1° to 20°, $M = 1.1$–2): the largest error is 0.46 %. The 20° cone at $M = 1.2$ has a detached shock and no conical solution: its half-angle is larger than the largest deflection an attached shock can produce at that Mach number. The grid convergence study (Figure 4b) is done on the Von Kármán nose at $L/D = 4$. At $M = 1.4$ it uses three grids ($80 \times 40$, $160 \times 80$ production, $320 \times 160$),

the observed order $p$ and the GCI of Roache ($F_s = 1.25$). At $M = 1.2$ the fine run did not finish in time, so the two-grid estimate is used: formal order $p = 2$ and $F_s = 3$.

The grid study quantifies the discretisation error. With the fine, production and coarse results $f_1$, $f_2$, $f_3$ and a refinement ratio $r = 2$ in each direction,

$$p = \frac{\ln\left[(f_3 - f_2)/(f_2 - f_1)\right]}{\ln r}, \qquad f_\infty = f_1 + \frac{f_1 - f_2}{r^p - 1}, \qquad \mathrm{GCI}_2 = F_s \left|\frac{f_2 - f_1}{f_2}\right| \frac{r^p}{r^p - 1}. \tag{5}$$

The observed order $p$ is the rate at which the error falls with the cell size. It should match the design order of the scheme (2), which shows that the grids are fine enough for the error to behave asymptotically. $f_\infty$ is the Richardson-extrapolated, grid-independent value. $\mathrm{GCI}_2$ is a conservative error band on the production-grid result, with the safety factor $F_s$. With two grids only, the band is $F_s|(f_3 - f_2)/f_2|/(r^p - 1)$, with the formal order $p = 2$ and the larger $F_s = 3$.

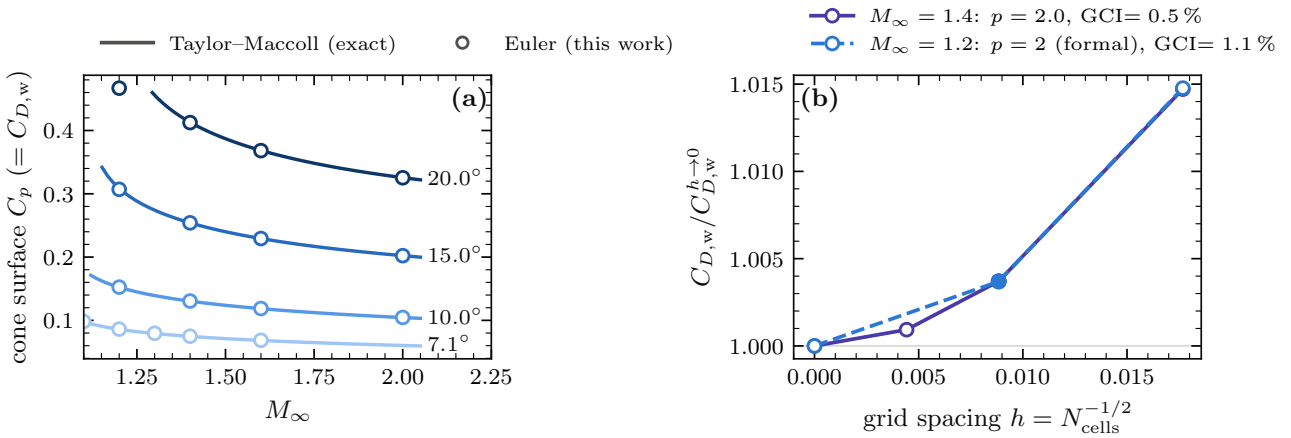



Figure 4: (a) Cone surface pressure from the Euler solver (markers) against the exact Taylor–Maccoll solution (lines). (b) Grid convergence of the nose wave drag, normalised by the Richardson-extrapolated value (open: computed grids; filled: production grid).

Figure 4a shows that the solver reproduces the exact cone pressures over the whole range of angles and Mach numbers, including the steep rise as the shock approaches detachment. Figure 4b gives $p = 2.0$ and a GCI of 0.5 % at $M = 1.4$ (1.1 % at $M = 1.2$ with the two-grid estimate). The production grid is therefore accurate to about 0.5–1 %, well below the 3–46 % by which the other shapes exceed the Von Kármán wave drag at $L/D = 4.5$ (Section 4).

## 3.2 Transonic range

Steady Euler solutions converge too slowly below $M = 1.1$ (detached shock far upstream, large subsonic region), and RANS is needed there anyway because the shock interacts with the boundary layer. The nose wave drag is therefore modelled as a smooth (cubic) rise from zero at $M = 0.8$ to the computed $M = 1.1$ value at $M = 1.0$, flat up to $M = 1.1$. This assumption is common to all shapes. The transonic Fluent runs listed in Section 9 replace it. Near $M = 1$, local supersonic pockets on the body end in shocks that thicken the boundary layer and can separate it, so viscous effects change the pressure field itself and the Euler model no longer applies. RANS (Reynolds-averaged Navier–Stokes) equations include the boundary layer with a turbulence model. The start of the rise at $M = 0.8$ corresponds to the first appearance of sonic flow on a slender nose. Because this model is the least certain part of the chain, its onset and amplitude are varied in the robustness study (Section 4.5).

## 3.3 Flight model

A point-mass trajectory is integrated from the 12 m rail ($84°$) to apogee. The integration is done in Cartesian coordinates; on the rail the direction is fixed, and after rail exit thrust and drag act along the velocity. Written along the flight path, with $V_\infty$ the speed and $\gamma_f$ the flight-path angle above the horizontal, the equations of motion are

$$m\frac{\mathrm{d}V_\infty}{\mathrm{d}t} = F_{\mathrm{th}} - \tfrac{1}{2}\rho V_\infty^2 A_b C_D - mg\sin\gamma_f, \qquad V_\infty\frac{\mathrm{d}\gamma_f}{\mathrm{d}t} = -g\cos\gamma_f, \tag{6}$$

with the mass $m$ decreasing linearly during the burn and the air density $\rho$ from the ISA at the current altitude. The first equation is Newton's second law along the velocity; the second describes the gravity turn, i.e. the slow tilting of the trajectory under gravity. The drag coefficient is built up from components (reference area $\pi D^2/4$):

$$C_D = k\,(C_{D,f} + C_{D,\mathrm{fins}} + C_{D,\mathrm{base}} + C_{D,e}) + C_{D,\mathrm{w}}(M), \tag{7}$$

with turbulent, roughness-limited skin friction on the wetted area of nose and body (Schlichting, compressibility-corrected), fin pressure drag (blunt leading edge subsonic, Ackeret hexagon supersonic), the OpenRocket base-drag law (nozzle exit area removed while burning) and the ERT-Sim excrescence law. The single factor $k$ is set so that the baseline nose (Von Kármán, 720 mm) reaches the LVO apogee of 9.0 km. The resulting $M_{\max} = 1.47$ is close to the LVO's Mach 1.4 without further tuning. A longer nose adds its length to the vehicle and its extra shell mass to the vehicle mass.

The terms of Equation (7) are the standard components of rocket drag estimation:

- $C_{D,f}$, *skin friction*: the shear stress of the turbulent boundary layer, proportional to the wetted area. It is computed from a flat-plate correlation as a function of the Reynolds number. When the viscous sublayer becomes thinner than the surface roughness, the friction coefficient stops decreasing with Reynolds number ("roughness-limited"). A correction accounts for the lower density in the hot boundary layer at high Mach number.
- $C_{D,\mathrm{fins}}$, *fin pressure drag*: from the blunt fin leading edges below Mach 1, and from linearised supersonic thin-airfoil theory (Ackeret) for the hexagonal section above it.
- $C_{D,\mathrm{base}}$, *base drag*: the low pressure in the wake behind the flat base. While the motor burns, the jet fills the nozzle exit area, which is then removed from the base area.
- $C_{D,e}$, *excrescences*: rail buttons, fasteners, joints and other protrusions, with the law of the ERT simulator.

Only $C_{D,\mathrm{w}}$ changes from one nose to the next. The length and mass effects of a longer nose enter through the wetted area in $C_{D,f}$ and through $m$.

## 3.4 Heating

The wall cannot exceed the stagnation temperature $T_0 = T_\infty(1 + \frac{\gamma-1}{2}M^2)$, which is the Firehorn design rule; the transient below shows how close it gets. $T_0$ follows from the steady energy equation $c_pT + \frac{1}{2}V^2 = \text{const}$ along a streamline: it is the temperature air reaches when it is brought to rest without heat exchange, all its kinetic energy being converted into enthalpy. For example, on the hot day ($T_\infty \approx 288$ K at 2.3 km) at $M = 1.56$, $T_0 = 288\times(1+0.2\times1.56^2) = 428$ K

(155 °C). How close a surface gets to this bound depends on the balance between the heat it receives, $q = h\,(T_r - T_w)$, and the heat it conducts away into the material. At the tip, the laminar stagnation-point coefficient (Fay–Riddell form without dissociation) with the modified-Newtonian velocity gradient is

$$h_s = 0.763\,\mathrm{Pr}^{-0.6}\sqrt{\rho_s \mu_s \left.\frac{\mathrm{d}u_e}{\mathrm{d}x}\right|_s}\,c_p, \qquad \left.\frac{\mathrm{d}u_e}{\mathrm{d}x}\right|_s = \frac{1}{r_n}\sqrt{\frac{2(p_s - p_\infty)}{\rho_s}}, \tag{8}$$

so $h_s \propto r_n^{-1/2}$. Here $\rho_s$, $\mu_s$ and $p_s$ are the density, viscosity and pressure at the stagnation point (behind the normal part of the bow shock), $c_p$ the specific heat of air and Pr its Prandtl number. $\mathrm{d}u_e/\mathrm{d}x|_s$ is the rate at which the flow at the edge of the boundary layer accelerates away from the stagnation point; the modified-Newtonian estimate gives it from the pressure difference and the tip radius. A larger tip radius gives a gentler acceleration and a thicker boundary layer, hence a lower heat-transfer coefficient. This is why re-entry vehicles have blunt noses. Along the shell, the turbulent flat-plate correlation $\mathrm{St} = 0.0296\,\mathrm{Re}_x^{*-0.2}\mathrm{Pr}^{-2/3}$ (Eckert reference temperature) drives the recovery temperature $T_r = T_\infty(1 + r\frac{\gamma-1}{2}M^2)$, $r = \mathrm{Pr}^{1/3}$. The Stanton number $\mathrm{St} = h/(\rho u c_p)$ is the dimensionless heat-transfer coefficient, and $\mathrm{Re}_x$ the Reynolds number on the distance $x$ from the tip. The asterisk means that the fluid properties are evaluated at Eckert's reference temperature, a weighted mean of the edge, wall and recovery temperatures that accounts for the variation of properties across a hot boundary layer. The recovery temperature is the temperature of an insulated wall under a boundary layer. It lies slightly below $T_0$ (recovery factor $r \approx 0.89$) because part of the dissipated energy is conducted away from the wall within the boundary layer. The wall response is a 1D transient conduction problem with an adiabatic inner face, which is conservative because it neglects radiation and internal convection:

$$\rho_m c_m \frac{\partial T}{\partial t} = k_m \frac{\partial^2 T}{\partial y^2}, \qquad -k_m \left.\frac{\partial T}{\partial y}\right|_{\text{outer}} = h\,(T_r - T_w), \qquad \left.\frac{\partial T}{\partial y}\right|_{\text{inner}} = 0, \tag{9}$$

with $\rho_m$, $c_m$, $k_m$ the density, specific heat and conductivity of the shell material and $y$ the coordinate through the thickness. Both are evaluated along the nominal and the hot-day envelope trajectories. A metal tip conducts heat so well that its temperature is nearly uniform; it is modelled as a single lumped mass heated through its wetted surface.

## 3.5 Stability, mass and payload

Barrowman's method gives the nose $C_{N\alpha} = 2$ at $x_{cp} = L - V/A_b$ and the fin contribution ($C_{N\alpha,f} = 7.4$ for the assumed fins). The change in static margin relative to the baseline nose is computed with positions measured from the aft end, including the CG shift from the nose-shell mass. The shell mass is the wetted area times the Firehorn as-built areal density (0.98 kg/m$^2$: 500 g measured for the 960 mm, Ø240 mm nose). The 3U fit is the length of curved nose over which $r \geq 77$ mm. Barrowman's method is the standard linear (small angle of attack, subsonic) method of model rocketry, also used by OpenRocket. It adds the normal-force slopes of the components and places the vehicle CP at their weighted mean,

$$x_{cp} = \frac{\sum_i C_{N\alpha,i}\,x_{cp,i}}{\sum_i C_{N\alpha,i}}, \qquad \text{static margin} = \frac{x_{cp} - x_{cg}}{D}, \tag{10}$$

with positions measured from the apex. Measuring from the aft end instead keeps the fins and the body fixed when the nose length changes.

### 3.6 Design catalogue, robustness and CAD

- **Direct runs at the design fineness.** All seven shapes (and the blunted Von Kármán nose) were also computed at $L/D = 4.5$ for $M = 1.6$, 1.4 and 1.2. The full flow fields were saved for the pressure maps of Section 5.
- **Robustness.** The apogee of every shape at $L/D = 4.5$ is recomputed with the onset of the transonic drag rise moved to $M = 0.75$ and 0.85, the transonic wave drag scaled by $\pm 20\,\%$, the rest of the vehicle drag increased by 20 %, and the hot-day envelope with the LVO masses. These perturbations target the least certain inputs: the transonic model, the calibration factor $k$ and the mass model.
- **Renders and CAD.** The shaded views of Figure 13 are computed from the exact profiles (Lambert–Phong shading of the surface of revolution). The STL solids are triangulated surfaces of revolution (128 azimuthal facets) closed by a flat base. Their closure and outward normals were checked, and their volume matches the analytical one.

## 4 Results

### 4.1 Nose wave drag

Figure 5a gives the wave drag of the seven profiles at $L/D = 4$ and Figure 6 the wall pressure behind it. The wall pressure is shown as the pressure coefficient $C_p = (p - p_\infty)/q_\infty$. By Equation (4), $C_{D,\mathrm{w}}$ is the integral of $C_p$ weighted by the forward-facing projected area of each surface element: regions of the nose with $C_p > 0$ add drag, and regions with $C_p < 0$ reduce it.

- **Von Kármán and power $n = \frac{1}{2}$ are the two low-drag profiles**, within a few per cent of each other: Von Kármán is lower at $M \geq 1.3$, power $\frac{1}{2}$ is lower at $M \leq 1.2$. They get there differently. The tangent profiles (Von Kármán, ogive, parabolic) expand continuously, and their wall pressure falls below ambient over the aft third. This partly cancels the overpressure near the tip. The power-$\frac{1}{2}$ nose concentrates its overpressure in a short spike at its blunt tip and stays just above ambient behind it. On a tangent profile the flow keeps expanding around the convex surface, and the aft part of the nose still faces slightly forward, so the suction there acts as a small thrust.
- **Ogive, parabolic and LV-Haack** carry 13–22 % more wave drag: their fuller forebody turns the flow harder near the tip. The larger surface angle over the front third compresses the flow more strongly, which shows as the higher $C_p$ of these shapes between $x/L \approx 0.1$ and 0.4 in Figure 6b.
- **The cone is the worst** at every Mach number, and its drag rises steeply towards Mach 1. Its surface pressure is the conical-flow value all the way to the shoulder. The surface angle is constant, so nothing along the nose relieves the overpressure, and the whole turn back to the axial direction happens in one expansion at the shoulder.
- **Fineness.** $C_{D,\mathrm{w}}$ falls as $1/(L/D)^2$ for every profile, the slope of slender-body theory (Figure 5b). For Von Kármán the Euler level is about 15 % below the linear-theory value $1/(L/D)^2$. Linear theory therefore reproduces the trend (Equation (2)) but not the level, which is why it is used here only to select the candidates and to interpolate between fineness ratios (Section 4.4).

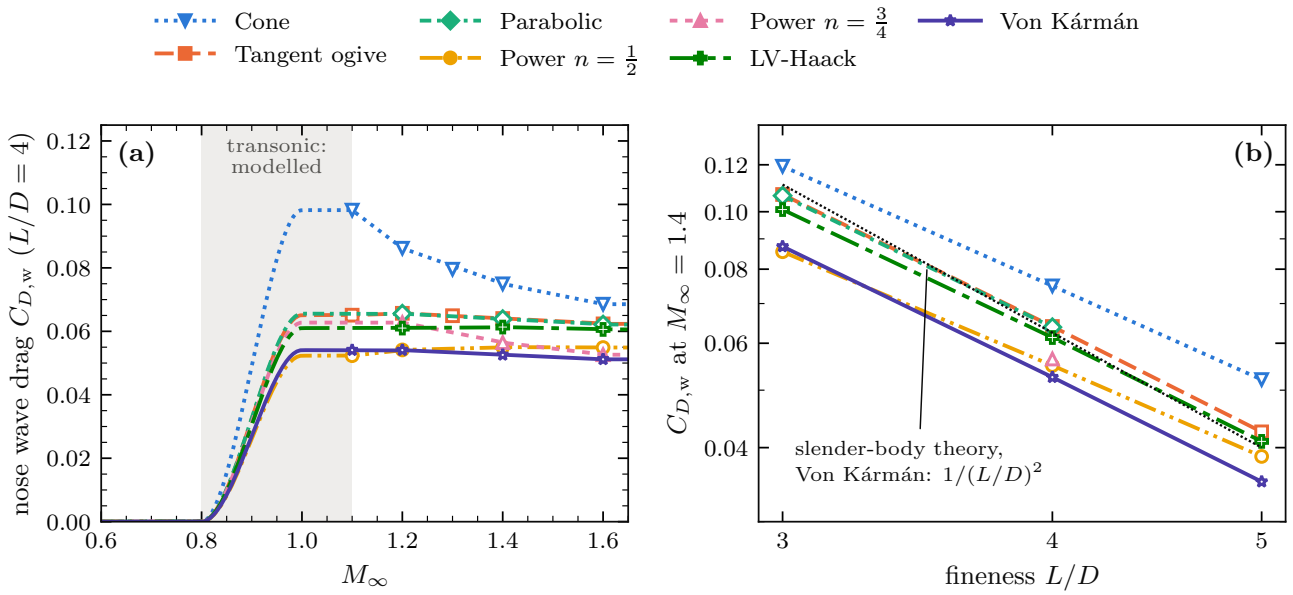



Figure 5: (a) Nose wave drag versus Mach number at $L/D = 4$ (markers: Euler; lines: model used in the trajectory; grey band: transonic model). (b) Wave drag versus fineness at $M = 1.4$, with the slender-body result for the Von Kármán nose.

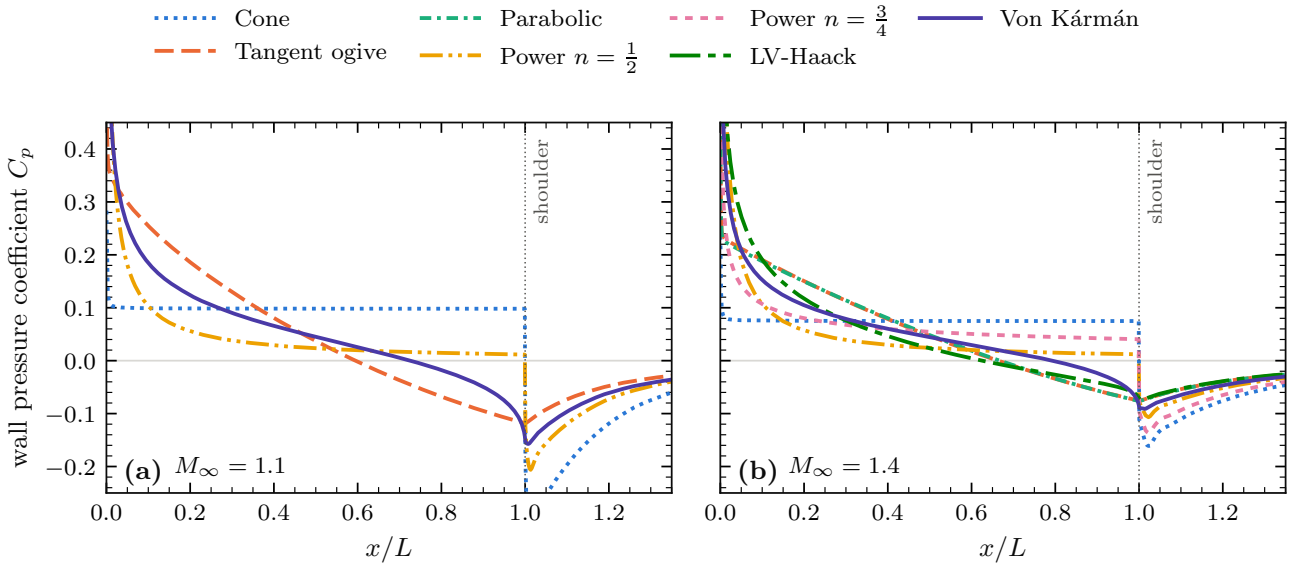



Figure 6: Wall pressure coefficient along the nose ($L/D = 4$) at (a) $M = 1.1$ (only the shapes computed at that Mach number) and (b) $M = 1.4$. The tangent profiles reach $C_p < 0$ over the aft third; the power-$\frac{1}{2}$ nose keeps a small overpressure up to the shoulder after a spike at its blunt tip; the cone keeps the conical value.

Figures 5 and 6 show that, at a given length, the drag is decided by the pressure distribution: shapes that compress the flow gently and then expand it along the rest of the nose have the lowest drag. They also show that length is as powerful as shape. Going from $L/D = 4$ to 4.5 lowers the Von Kármán wave drag at $M = 1.4$ by 19 % (0.0526 to 0.0425), close to the factor $(4/4.5)^2 = 0.79$ of Equation (2); at equal length, the ogive carries 21 % more than the Von Kármán nose. The design question is therefore how to combine the best profile with the longest nose the vehicle can accommodate.

## 4.2 Flight performance

Figure 7a shows the nominal and envelope ascents. Weisshorn burns out at 9.1 s near 2.2 km at $M_{\max} = 1.47$ (nominal) or 1.56 (envelope). It spends 16 s above $M = 0.8$ and 10 s above Mach 1. Figure 7b shows where the drag impulse $\int D\,\mathrm{d}t$ is spent: mostly between $M = 0.8$ and 1.5, which is exactly where the nose wave drag acts. The apogee sensitivities around the baseline are **–100 m per** $\Delta C_D = 0.01$ and **–143 m/kg**.

The vehicle accelerates during the burn and then decelerates under drag and gravity. It crosses the transonic band twice, once accelerating and once coasting (Figure 7a). The drag impulse, the integral of the drag force over time, is the momentum the vehicle loses to drag. It is concentrated at high Mach number because the dynamic pressure is highest there (high speed at low altitude), which is why a drag saving above $M = 0.8$ is worth so much apogee. The two sensitivities convert any drag or mass difference into apogee. The $C_D$ sensitivity applies to a change over the whole flight. A wave-drag difference only acts above $M = 0.8$ and is therefore worth somewhat less: the ogive carries 0.009 more $C_{D,\mathrm{w}}$ than Von Kármán at $M = 1.4$ and $L/D = 4.5$, and loses 79 m rather than 90 m.

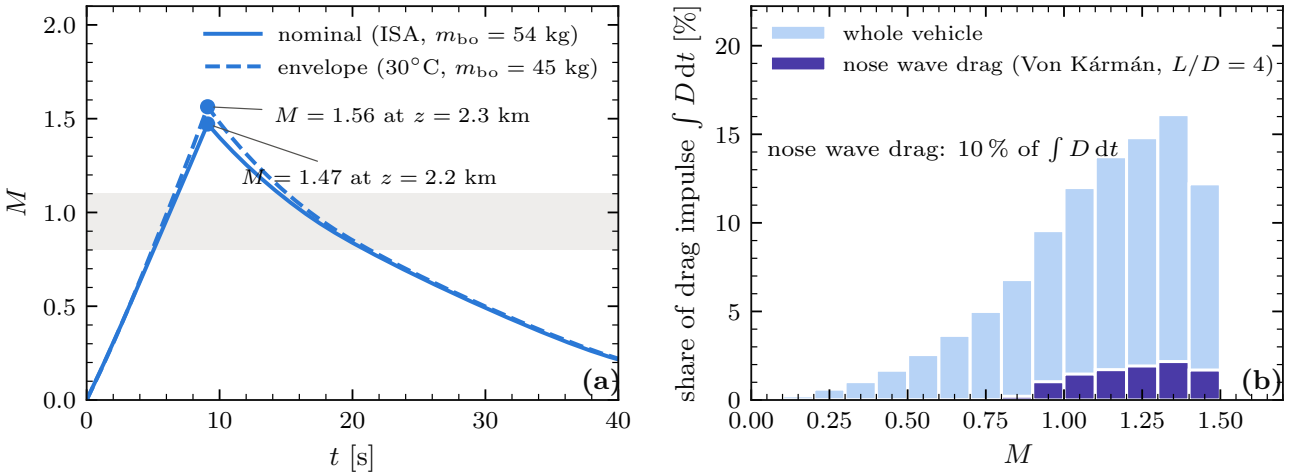



Figure 7: (a) Mach number along the nominal and hot-day envelope ascents (grey band: transonic range). (b) Share of the drag impulse per Mach bin for the baseline nose; the dark part is the nose wave drag.

Figure 8 breaks the vehicle drag down by component for the recommended nose. Skin friction dominates at all speeds (Weisshorn is a fineness-30 airframe). Base drag is the second contributor, and the nose wave drag appears above $M = 0.8$, where it is 10–12 % of the total. This is why the nose shape moves the apogee by tens of metres rather than kilometres, and why the length (wetted area) and the shape effects are of the same order. A fineness-30 airframe (5.4 m long for 0.18 m in diameter) has a large wetted area for its frontal area, so friction outweighs the pressure drag terms. The total drag coefficient peaks just above $M = 1$, where the base, fin and nose pressure drags are all near their maximum.

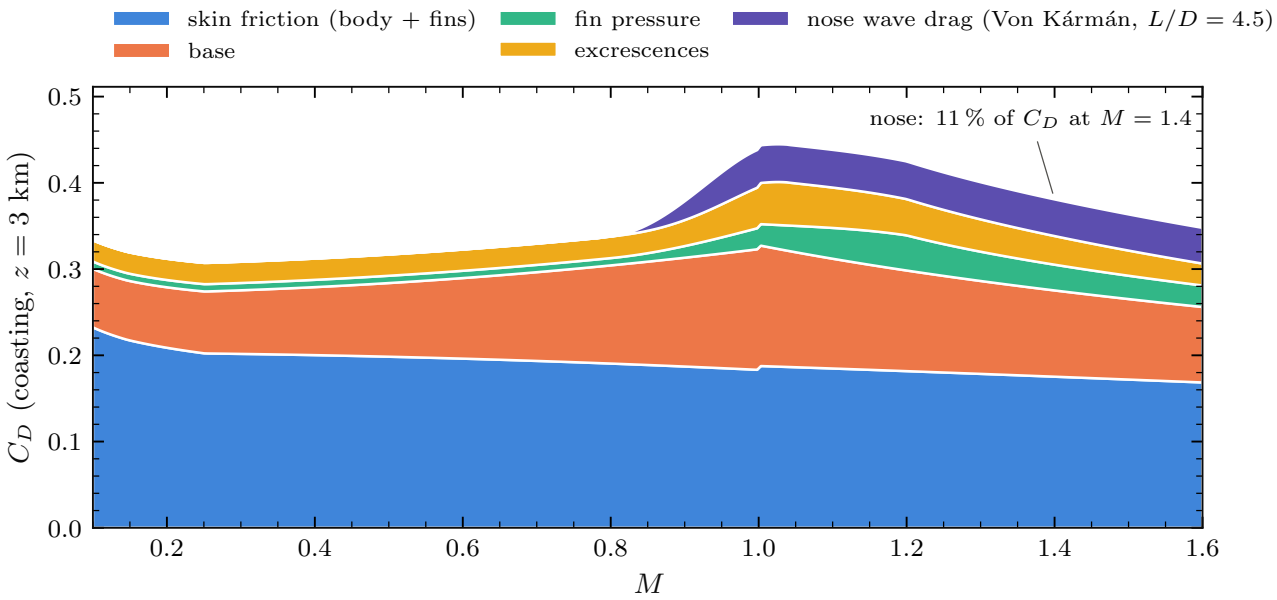



Figure 8: Drag build-up of Weisshorn with the recommended nose (coasting, $z = 3$ km, rest of the vehicle scaled by the calibration factor $k$). The kink at $M \approx 0.25$ is where the skin friction becomes roughness-limited; the one at $M = 1$ is the switch of the base-drag law.

## 4.3 Trade study

Figure 9 gives, for every profile and fineness, the apogee change relative to the baseline, the length over which a 3U section fits, and the change in static margin. Table 3 lists the candidates closest to the LVS length. Each point of Figure 9a is a full trajectory with the wave drag of that nose, its own length, wetted area and shell mass. The curves rise with fineness because the wave drag falls as $1/(L/D)^2$. They flatten because the friction and mass of the longer nose grow linearly with $L$.

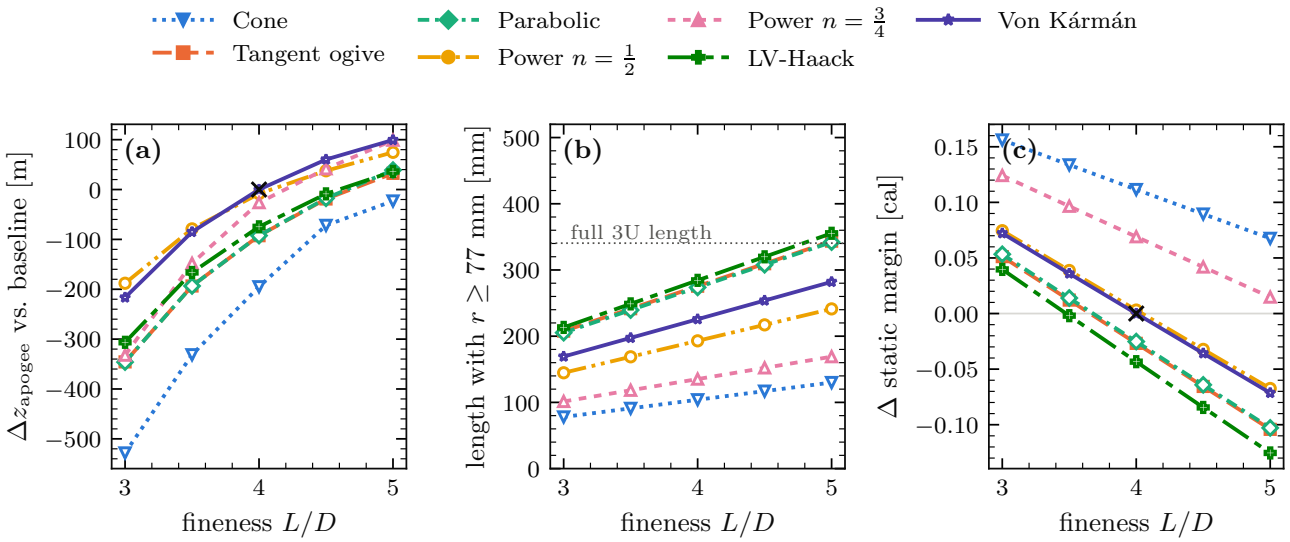



Figure 9: (a) Apogee change relative to the baseline (Von Kármán, $L/D = 4$, cross), including the extra length, wetted area and mass of longer noses. (b) Length of the curved nose with $r \geq 77$ mm (3U fit). (c) Change in subsonic static margin.

Table 3: Candidates at $L/D = 4$ (LVS length, 720 mm) and $L/D = 4.5$ (810 mm). $\Delta z$: apogee change vs. the baseline (Von Kármán, $L/D = 4$); mass: shell + Al tip; 3U in nose: curved length with $r \geq 77$ mm; bay: curved length plus the cylinder the 3U still needs.

| $L/D$ | Profile | $C_{D,\mathrm{w}}$ at $M =$ | | | $\Delta z$ | mass | 3U in nose | bay | $\Delta$SM |
|---|---|---|---|---|---|---|---|---|---|
| | | 1.1 | 1.2 | 1.4 | [m] | [g] | [mm] | [mm] | [cal] |
| 4 | Cone | 0.0982 | 0.0862 | 0.0750 | -195 | 261 | 104 | 966 | +0.11 |
| 4 | Tangent ogive | 0.0651 | 0.0657 | 0.0640 | -93 | 328 | 275 | 795 | -0.03 |
| 4 | Parabolic | 0.0656 | 0.0656 | 0.0639 | -92 | 328 | 274 | 797 | -0.03 |
| 4 | Power $n = \frac{1}{2}$ | 0.0523 | 0.0541 | 0.0550 | -8 | 328 | 193 | 878 | +0.00 |
| 4 | Power $n = \frac{3}{4}$ | 0.0628 | 0.0628 | 0.0564 | -26 | 289 | 135 | 935 | +0.07 |
| 4 | LV-Haack | 0.0610 | 0.0610 | 0.0613 | -75 | 341 | 284 | 786 | -0.04 |
| 4 | **Von Kármán** | 0.0541 | 0.0541 | 0.0526 | +0 | 321 | 226 | 845 | +0.00 |
| 4.5 | Cone | 0.0708 | 0.0708 | 0.0619 | -72 | 286 | 117 | 1043 | +0.09 |
| 4.5 | Tangent ogive | 0.0530 | 0.0530 | 0.0516 | -19 | 361 | 309 | 870 | -0.07 |
| 4.5 | Parabolic | 0.0529 | 0.0529 | 0.0515 | -18 | 361 | 308 | 870 | -0.06 |
| 4.5 | Power $n = \frac{1}{2}$ | 0.0450 | 0.0450 | 0.0457 | +38 | 361 | 217 | 943 | -0.03 |
| 4.5 | Power $n = \frac{3}{4}$ | 0.0513 | 0.0513 | 0.0463 | +42 | 318 | 152 | 1008 | +0.04 |
| 4.5 | LV-Haack | 0.0496 | 0.0496 | 0.0497 | -9 | 376 | 320 | 870 | -0.08 |
| 4.5 | **Von Kármán** | 0.0437 | 0.0437 | 0.0425 | +60 | 354 | 254 | 907 | -0.04 |

- **Von Kármán, power $\frac{1}{2}$ and power $\frac{3}{4}$ lead at every length** and swap places within about 30 m. Power $\frac{1}{2}$ is ahead at $L/D \leq 3.5$, Von Kármán at $L/D = 4$–4.5, and the power-$\frac{3}{4}$ estimate (scaled from $L/D = 4$) at $L/D = 5$. At $L/D = 4$, Von Kármán beats the tangent ogive (Bella Lui II / Nordend) by 93 m and the Firehorn power-$\frac{1}{2}$ by 8 m.
- **Length pays, with diminishing returns**: 60 m per extra 90 mm ($\Delta L/D = 0.5$) around $L/D = 4$–4.5.
- **Fuller noses house more payload**: at $L/D = 4$ the ogive and LV-Haack fit 275 mm to 284 mm of the 3U inside the curved part, against 226 mm for Von Kármán. Measured as nose-bay length (curved part + the cylinder still needed for the 3U), the Von Kármán nose still gives the highest apogee per millimetre of bay length. In Figure 9b only the fullest noses at $L/D = 5$ approach the full 3U length (dotted line).
- **Stability is not a discriminator**: all candidates change the static margin by less than 0.1 cal. A longer nose lowers the margin slightly (Figure 9c): its CP, a distance $V/A_b$ ahead of the shoulder, moves forward with $L$ faster than its extra mass moves the CG. The largest change over the whole design space, 0.16 cal, is small against the 2–4 cal target.

The trade study thus separates two groups: the low-drag group (Von Kármán and the power noses) and the high-volume group (ogive, parabola, LV-Haack). Section 6 weighs drag against payload room and manufacturing.

## 4.4 Check at the design fineness

The trade study interpolates the Euler results between $L/D = 3$, 4 and 5 (on $C_{D,\mathrm{w}}(L/D)^2$, which is nearly constant for slender noses). To check this, every shape was also computed directly at $L/D = 4.5$ (Table 4). Where three fineness ratios were available the interpolation is within 0.3 %. The parabolic and power-$\frac{3}{4}$ noses, first computed at $L/D = 4$ only, were underestimated

by 2–4 %. The direct values are now used everywhere, which moves the power-$\frac{3}{4}$ nose from $+55$ to $+42$ m at $L/D = 4.5$. The product $C_{D,\mathrm{w}}(L/D)^2$ is nearly constant because of Equation (2). Interpolating it, rather than $C_{D,\mathrm{w}}$ itself, removes most of the fineness dependence and leaves a slowly varying function. Scaling from a single fineness assumes that the $1/(L/D)^2$ law holds exactly, and the 2–4 % error is the departure from it.

Table 4: Wave drag at $L/D = 4.5$: value used before the direct runs (interpolated in $C_{D,\mathrm{w}}(L/D)^2$, or scaled with $1/(L/D)^2$ from a single fineness) against the direct Euler result, and the error of the former.

| | | $M = 1.2$ | | | $M = 1.4$ | | | $M = 1.6$ | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Shape | source | interp. | direct | [%] | interp. | direct | [%] | interp. | direct | [%] |
| Cone | interp. 3–5 | 0.0707 | 0.0708 | -0.2 | 0.0618 | 0.0619 | -0.2 | 0.0566 | 0.0567 | -0.2 |
| Tangent ogive | interp. 3–5 | 0.0528 | 0.0530 | -0.2 | 0.0515 | 0.0516 | -0.2 | 0.0503 | 0.0504 | -0.2 |
| Parabolic | interp. 4–5 | 0.0518 | 0.0529 | -2.0 | 0.0505 | 0.0515 | -2.1 | 0.0491 | 0.0502 | -2.3 |
| Power $n = \frac{1}{2}$ | interp. 3–5 | 0.0449 | 0.0450 | -0.2 | 0.0456 | 0.0457 | -0.3 | 0.0455 | 0.0456 | -0.3 |
| Power $n = \frac{3}{4}$ | scaled from 4 | 0.0496 | 0.0513 | -3.4 | 0.0446 | 0.0463 | -3.6 | 0.0416 | 0.0432 | -3.7 |
| LV-Haack | interp. 3–5 | 0.0494 | 0.0496 | -0.3 | 0.0495 | 0.0497 | -0.3 | 0.0490 | 0.0491 | -0.2 |
| Von Kármán | interp. 3–5 | 0.0436 | 0.0437 | -0.2 | 0.0424 | 0.0425 | -0.2 | 0.0413 | 0.0414 | -0.2 |

## 4.5 Robustness of the ranking

The weakest parts of the chain are the transonic drag model, the drag of the rest of the vehicle and the mass model. A ranking is only useful if it does not depend on assumptions that cannot yet be verified. Figure 10 recomputes the apogee of every shape at $L/D = 4.5$ under six perturbations of these assumptions and shows it relative to Von Kármán in the same scenario:

- **Von Kármán flies highest in every scenario.** The power noses stay within 31 m of it (11 m to 31 m). The ogive, parabolic and LV-Haack noses are 57 m to 98 m lower, and the cone 98 m to 156 m lower.
- The spread between scenarios is smaller than the gap between the groups, so the selection does not hinge on the transonic assumption. The largest effect is the hot-day envelope with the LVO masses, which raises the Mach number and amplifies the differences.
- A draggier vehicle (+20 % on the rest of the drag) shrinks the differences: the nose matters less when it is a smaller share of the total.

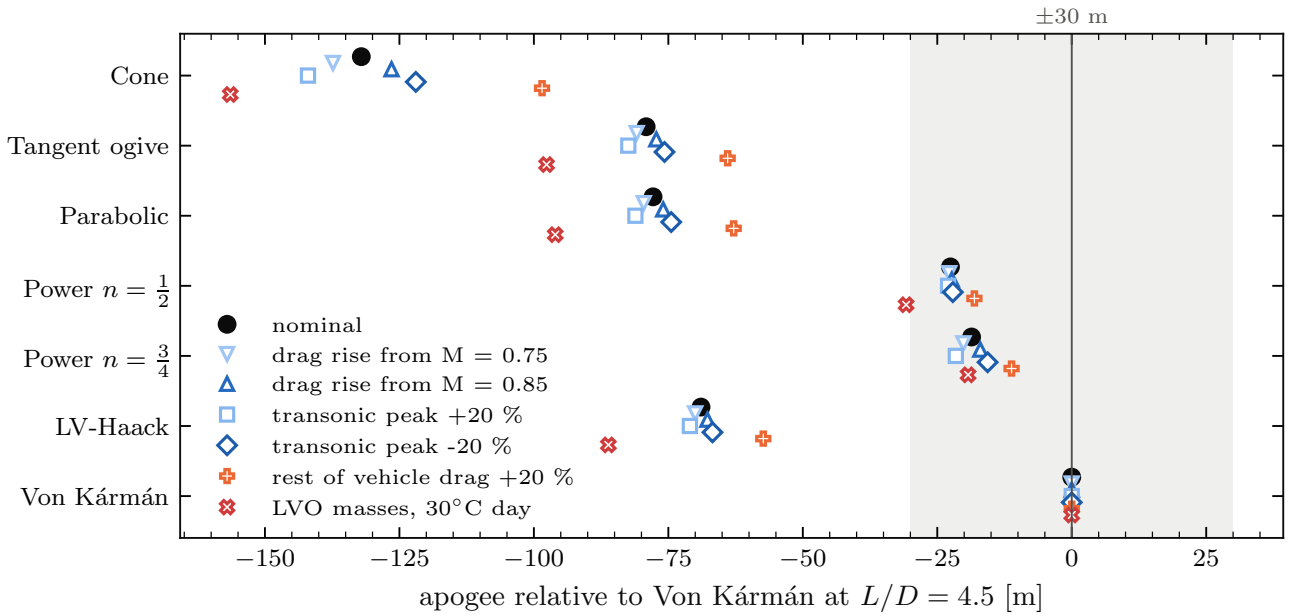



Figure 10: Apogee of each design at $L/D = 4.5$ relative to Von Kármán, under the nominal model and six perturbed ones. The grey band is ±30 m, about ±0.2 kg of dry mass.

Each row of Figure 10 is one design; the markers of a row are its apogee deficit under the different models. The grey band marks differences of the order of the mass uncertainty of a vehicle at this stage (0.2 kg); designs inside it are equivalent in practice. Only the two power noses lie in it or at its edge, and no design crosses zero.

## 4.6 Tip bluntness

A sharp tip cannot be laid up and burns or chips first, so the tip is a separate machined part. Figure 11a gives the drag cost of a spherical tip on the Von Kármán nose. At $M = 1.4$ a 4.5 mm tip ($r_n/R = 0.05$) costs $+0.6\,\%$ (–2 m of apogee), and a 9 mm tip ($r_n/R = 0.1$) costs $+3.7\,\%$ (–10 m). The cost grows quickly with radius because the sphere replaces the slender front of the Haack profile. Figure 11b gives the benefit: the stagnation-point heat-transfer coefficient falls as $r_n^{-1/2}$. The sphere is tangent to the profile and carries a small detached bow shock (Section 2.1). At its centre the pressure is the stagnation pressure behind a normal shock, $C_p \approx 1.5$ at $M = 1.4$, against about 0.04 averaged over a sharp nose. The penalty rises with Mach number because this stagnation pressure does. For the selected 4.5 mm tip, $h_s$ is about 2 kW/(m$^2$ K); doubling the radius would only lower it by $\sqrt{2}$. A radius of 4.5 mm therefore keeps the drag cost below 1 %, and the heating is handled by the choice of tip material (Section 4.7).

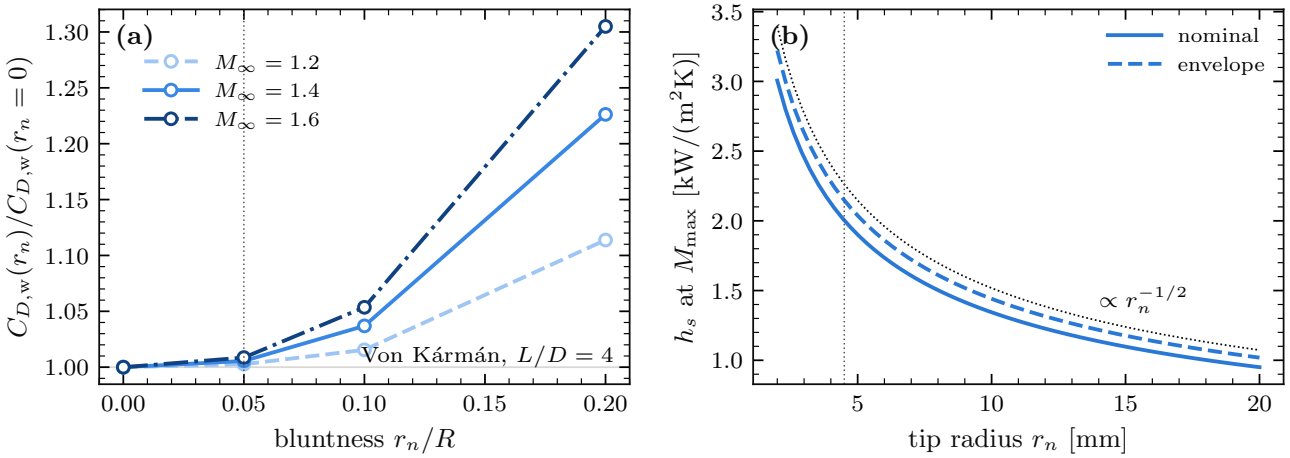



Figure 11: (a) Wave drag of the blunted Von Kármán nose relative to the sharp one. (b) Stagnation-point heat-transfer coefficient at $M_{\max}$ versus tip radius (dotted line: the selected 4.5 mm).

## 4.7 Heating

Figure 12 gives the temperature histories. The peak stagnation temperature is 119 °C (nominal) and 156 °C (envelope), reached at burn-out.

- **Tip.** A 3D-printed PET-CF tip (Firehorn) has a low conductivity, so its surface follows $T_0$ and reaches 104 °C (nominal) and 138 °C (envelope). That is well above the 70 °C to 80 °C heat-deflection range of PET/PETG-CF. A 60 g Al 6082-T6 tip spreads the heat and stays below 51 °C (nominal) and 73 °C (envelope). In the model PET-CF conducts heat 600 times less well than aluminium (0.3 against 180 W/(m K)). The heat received at the surface of a printed tip stays in a thin layer, whose temperature quickly approaches the recovery temperature. In the aluminium tip the same heat spreads through the whole 60 g, whose heat capacity limits the temperature rise. The heat-deflection temperature (HDT) is the temperature at which a standard polymer specimen deforms by a set amount under a set load; above it a printed tip would soften.
- **Shell.** The 1.2 mm CFRP shell follows the recovery temperature within a few seconds. Its inner face peaks at 77 °C (nominal) and 104 °C (envelope), about 3 s after burn-out, then cools. This is below the cure temperature of the prepreg, but the envelope case comes close to the DP490 coupler bond (shear strength 2.8 MPa at 120 °C). The Firehorn thermal requirement (390 K for 5 s) does not cover the envelope; Section 9 proposes 430 K for 10 s. The inner face peaks after burn-out because heat needs time to diffuse through the laminate: the outer surface follows the recovery temperature, while the inner face lags behind it and keeps warming for a few seconds after the outer surface has started to cool.

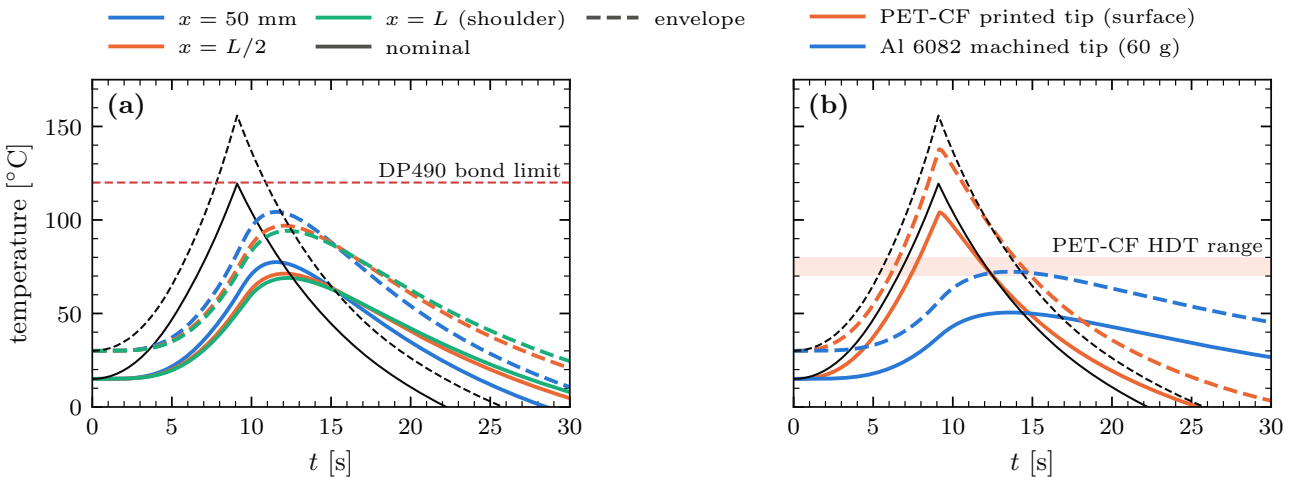


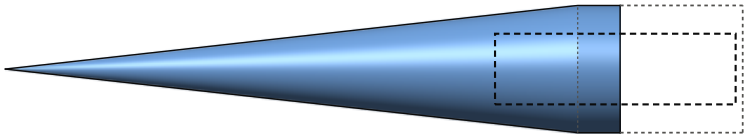

Figure 12: (a) Inner-face temperature of the CFRP shell at three stations (adiabatic inner face, conservative). (b) Tip temperature: printed PET-CF surface against a machined Al 6082 insert. Solid: nominal ascent; dashed: hot-day envelope; black: stagnation temperature $T_0$.

Figure 12 shows that heating is a material question, not a shape question. The stagnation temperature is set by the trajectory, the same for every profile. What the design controls is how much of it reaches a temperature-sensitive part, the tip polymer or the adhesive bond. Both are protected by the choices made here: a metal tip and a thermal requirement that covers the hot-day envelope.

# 5 Design Catalogue

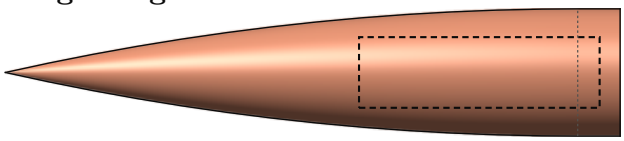
This section presents the seven candidate nose cones as designs at the recommended length, $L/D = 4.5$ ($L = 810$ mm, $D = 180$ mm), each with the 60 mm bonded shoulder of the Firehorn design. Figure 13 shows them to scale with the 3U payload placed as far forward as each profile allows, and Figure 14 shows the computed flow around each of them at $M = 1.4$. Table 5 collects all the numbers. The wave-drag values in this section come from Euler runs made directly at $L/D = 4.5$; they agree with the values interpolated from $L/D = 3$, 4, 5 to within 0.3 % (Section 4.4). Watertight STL solids of every design are in `results/cad/` (Appendix A). The catalogue turns the trade study into concrete hardware: for each shape it shows the outer mould line that would be built, where the payload sits in it, and the flow it produces.

**Cone**



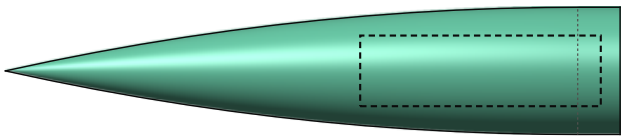
$\eta = \xi$
$C_{D,\mathrm{w}}(1.4) = 0.0619$, $\Delta z = -72$ m
mass 286 g, 3U in nose 117 mm
shoulder $6.3^\circ$, $x_{cp}/L = 0.67$

**Tangent ogive**



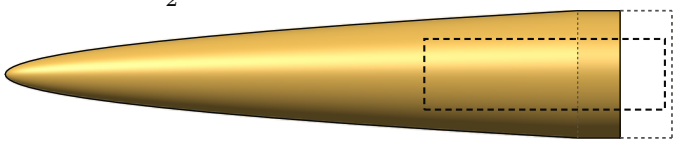
circular arc, $\rho = (R^2 + L^2)/2R$
$C_{D,\mathrm{w}}(1.4) = 0.0516$, $\Delta z = -19$ m
mass 361 g, 3U in nose 309 mm
shoulder $0.0^\circ$, $x_{cp}/L = 0.46$

**Parabolic**



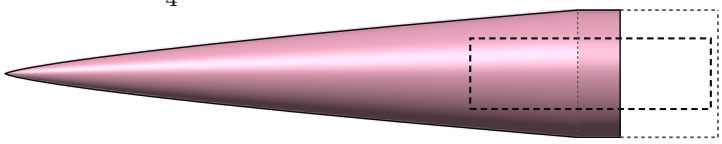
$\eta = 2\xi - \xi^2$
$C_{D,\mathrm{w}}(1.4) = 0.0515$, $\Delta z = -18$ m
mass 361 g, 3U in nose 308 mm
shoulder $0.0^\circ$, $x_{cp}/L = 0.47$

**Power $n = \frac{1}{2}$**



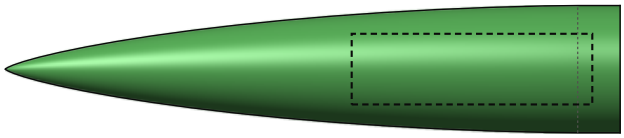
$\eta = \xi^{1/2}$
$C_{D,\mathrm{w}}(1.4) = 0.0457$, $\Delta z = +38$ m
mass 361 g, 3U in nose 217 mm
shoulder $3.2^\circ$, $x_{cp}/L = 0.50$

**Power $n = \frac{3}{4}$**



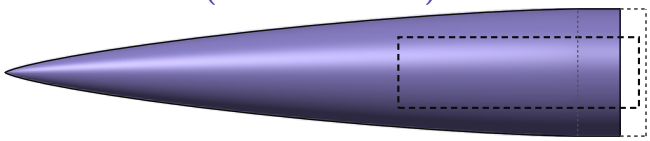
$\eta = \xi^{3/4}$
$C_{D,\mathrm{w}}(1.4) = 0.0463$, $\Delta z = +42$ m
mass 318 g, 3U in nose 152 mm
shoulder $4.8^\circ$, $x_{cp}/L = 0.60$

**LV-Haack**



Haack, $C = \frac{1}{3}$
$C_{D,\mathrm{w}}(1.4) = 0.0497$, $\Delta z = -9$ m
mass 376 g, 3U in nose 320 mm
shoulder $0.0^\circ$, $x_{cp}/L = 0.44$

**Von Kármán (recommended)**



Haack, $C = 0$
$C_{D,\mathrm{w}}(1.4) = 0.0425$, $\Delta z = +60$ m
mass 354 g, 3U in nose 254 mm
shoulder $0.0^\circ$, $x_{cp}/L = 0.50$

Figure 13: The seven designs at $L/D = 4.5$, rendered to the same scale (diameter 180 mm). Thin dotted line: end of the curved part; dashed box: 3U CubeSat ($100 \times 100 \times 340.5$ mm) pushed forward until its corners reach $r = 77$ mm; dotted extension: payload-bay cylinder still needed behind the 60 mm shoulder. $\Delta z$: apogee change relative to the LVS-length baseline (Von Kármán at $L/D = 4$).

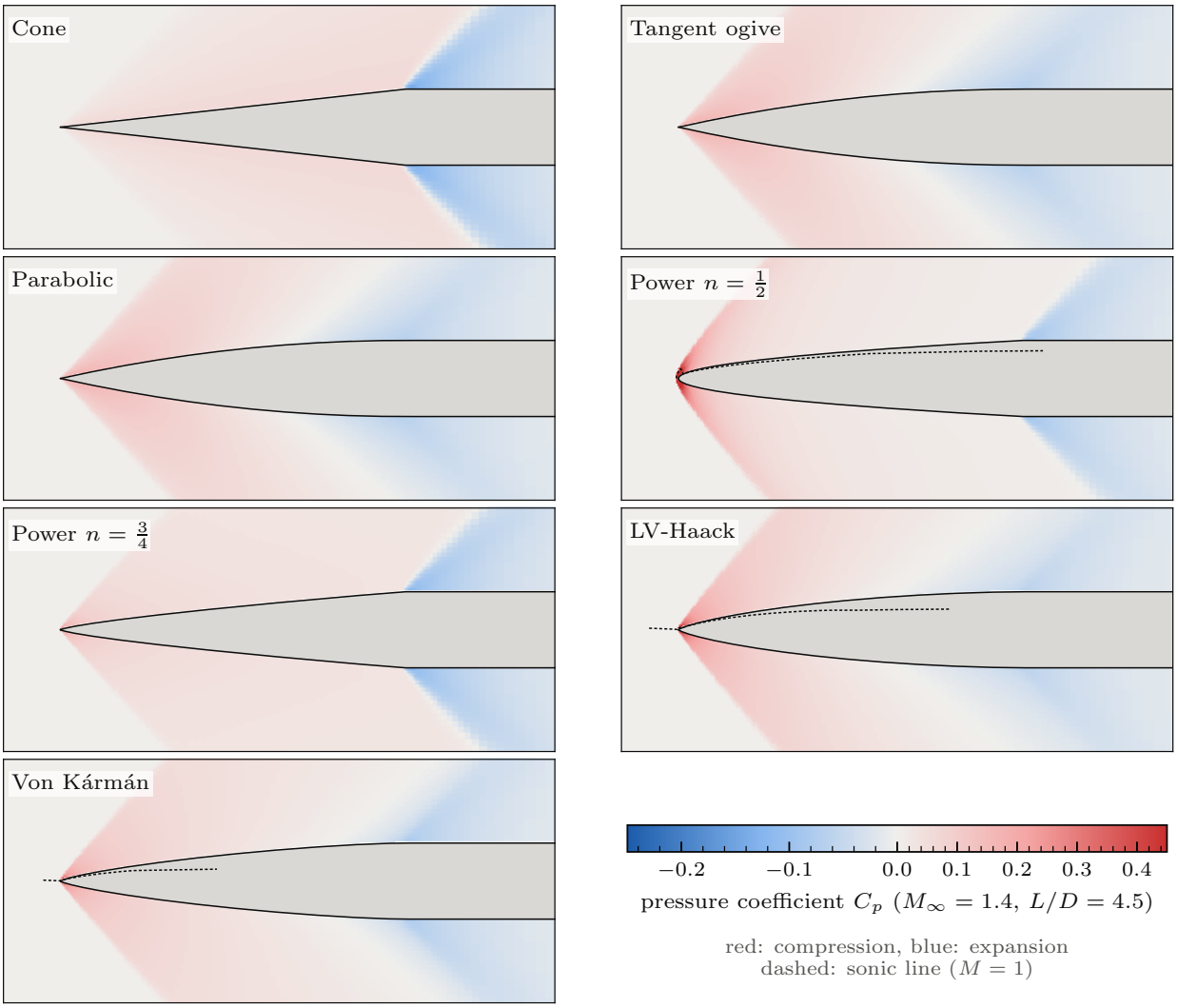



Figure 14: Pressure coefficient around the seven designs at $M_\infty = 1.4$, $L/D = 4.5$ (axisymmetric Euler, mirrored about the axis). The bow shock is the edge of the red region; the blue fans are the expansions that turn the flow back along the cylinder. The dashed line marks the sonic line where a subsonic region exists.

In Figure 13, the further forward the dashed 3U box can be pushed, the shorter the payload-bay cylinder behind the shoulder (dotted extension). In Figure 14, the red region ahead of each nose is the air compressed between the bow shock and the body; its outer edge, the shock, leans back at roughly the Mach angle ($45.6°$ at $M = 1.4$, Equation (1)). The blue regions are expansions. On the cone and the power noses they are concentrated in a fan at the shoulder kink; on the tangent profiles they are spread over the aft part of the nose itself. The more intense the red region in contact with the nose surface, and the less blue on it, the higher the wave drag.

Table 5: Design catalogue at $L/D = 4.5$ ($L = 810$ mm). Wave drag from the direct Euler runs at this fineness; $\Delta z$ relative to Von Kármán at $L/D = 4$ (LVS length), including length and mass effects.

| | Cone | Ogive | Parabolic | Power $\frac{1}{2}$ | Power $\frac{3}{4}$ | LV-Haack | Von Kármán |
|---|---|---|---|---|---|---|---|
| Volume [L] | 6.87 | 11.03 | 10.99 | 10.31 | 8.24 | 11.59 | 10.31 |
| Wetted area [m$^2$] | 0.230 | 0.308 | 0.307 | 0.307 | 0.263 | 0.322 | 0.300 |
| Tip semi-angle [°] | 6 | 13 | 13 | 89 | 74 | 80 | 78 |
| Shoulder angle [°] | 6.3 | 0.0 | 0.0 | 3.2 | 4.8 | 0.0 | 0.0 |
| $x_{cp}/L$ (Barrowman) | 0.667 | 0.465 | 0.467 | 0.500 | 0.600 | 0.438 | 0.500 |
| $C_{D,\mathrm{w}}$ at $M = 1.2$ | 0.0708 | 0.0530 | 0.0529 | 0.0450 | 0.0513 | 0.0496 | 0.0437 |
| $C_{D,\mathrm{w}}$ at $M = 1.4$ | 0.0619 | 0.0516 | 0.0515 | 0.0457 | 0.0463 | 0.0497 | 0.0425 |
| $C_{D,\mathrm{w}}$ at $M = 1.6$ | 0.0567 | 0.0504 | 0.0502 | 0.0456 | 0.0432 | 0.0491 | 0.0414 |
| $\Delta z$ at $L/D = 4$ [m] | -195 | -93 | -92 | -8 | -26 | -75 | +0 |
| $\Delta z$ at $L/D = 4.5$ [m] | -72 | -19 | -18 | +38 | +42 | -9 | +60 |
| $\Delta z$ at $L/D = 5$ [m] | -23 | +32 | +39 | +75 | +99 | +36 | +99 |
| Shell + tip mass [g] | 286 | 361 | 361 | 361 | 318 | 376 | 354 |
| 3U inside nose [mm] | 117 | 309 | 308 | 217 | 152 | 320 | 254 |
| Nose-bay length [mm] | 1043 | 870 | 870 | 943 | 1008 | 870 | 907 |
| $\Delta$ static margin [cal] | +0.09 | -0.07 | -0.06 | -0.03 | +0.04 | -0.08 | -0.04 |

Table 5 groups the metrics in three blocks: geometry (volume, wetted area, angles, centre of pressure), aerodynamics and flight (wave drag, apogee change at three lengths), and integration (mass, payload room, bay length, static margin). The datasheets below discuss each design in the same order.

## 5.1 Conical

**Geometry.** A straight generator, $6.3°$ semi-angle at $L/D = 4.5$. It has the smallest volume (6.9 L) and wetted area (0.230 m$^2$) of all designs, hence the lightest shell (286 g), but it meets the cylinder with a $6.3°$ kink.

**Flow.** The conical-flow pressure ($C_p \approx 0.06$ at $M = 1.4$) acts on the whole surface without relaxing. All the turning happens at once in a strong expansion fan at the shoulder (Figure 14). That is why it has the highest wave drag at every Mach number, 46 % above Von Kármán at $M = 1.4$, and why its drag climbs steeply towards Mach 1. As $M$ decreases, the conical shock strengthens and moves towards detachment, so the uniform surface pressure rises (Figure 4a).

**Integration and manufacture.** Only 117 mm of the 3U fits inside the cone, so the nose bay grows to 1043 mm. The mould is the simplest of all (a developable surface, i.e. one that can be unrolled onto a plane without stretching, so it can even be laid up from flat patterns), but the shoulder kink needs a fillet at the coupler joint.

**Verdict: rejected.** It is the worst on drag ($-132$ m against Von Kármán), payload room and bay length. It also has the most aft centre of pressure ($x_{cp}/L = 2/3$), which is helpful for stability but irrelevant here because the margin changes by only $+0.09$ cal. (An aft nose CP moves the vehicle CP back, which increases the static margin.)

## 5.2 Tangent ogive

**Geometry.** A circular arc tangent to the body, the classic subsonic nose used on Bella Lui II and Nordend. It has a $13°$ tip semi-angle, 11.0 L of volume and a tangent shoulder. Its sharp apex has a finite angle, so near the tip it behaves like a $13°$ cone, about twice the semi-angle of the $6.3°$ cone of the same length.

**Flow.** The fuller forebody turns the flow harder near the tip (stronger compression than Von Kármán), then expands continuously; the wall pressure goes below ambient over the aft 40 %. Wave drag is 21 % above Von Kármán at $M = 1.4$.

**Integration and manufacture.** It is the best of the low-drag families for packaging: 309 mm of the 3U fits in the curved part, giving a 870 mm nose bay. The constant curvature makes it the easiest smooth profile to template or machine: a single radius describes the whole generator.

**Verdict: acceptable alternative for volume-driven designs.** It costs 79 m of apogee against Von Kármán, about 0.55 kg of dry-mass margin, for 37 mm less nose bay.

## 5.3 Parabolic ($K' = 1$)

**Geometry and flow.** At the same length the parabola is practically the tangent ogive. The volume, wetted area, wave drag and pressure field are all the same (wave drag within 0.4 %), and its shoulder is also tangent.

**Verdict: same as the ogive**, without the ERT heritage. It is listed for completeness: it is the "parabola" that Crowell's chart rates best in the high subsonic range (Table 6, Section 5.8), which is irrelevant for a vehicle whose drag losses are concentrated at $M = 0.8$–1.5.

## 5.4 Power series $n = \frac{1}{2}$ (Firehorn)

**Geometry.** $\eta = \xi^{1/2}$ starts with a vertical tangent: the tip is blunt by construction. It has the same volume as Von Kármán (10.3 L), but the volume sits further forward, and it meets the cylinder with a $3.2°$ kink.

**Flow.** The blunt apex detaches the bow shock and creates a small subsonic pocket (the only closed sonic line in Figure 14). The sonic line encloses the region where the flow behind the normal part of the shock has been slowed below Mach 1; the pressure there is close to the stagnation pressure, which gives the spike of Figure 6. The overpressure is a short, intense spike at the apex; behind it the wall pressure stays slightly above ambient all the way to the shoulder, where the kink produces an expansion. The net wave drag is only 3–10 % above Von Kármán at $L/D = 4.5$. On short noses ($L/D = 3$), and at $M \leq 1.1$ for $L/D = 4$, it is the lowest of all. Its drag is concentrated on the small frontal area of the blunt apex, whereas the rest of the nose is very slender and carries almost no overpressure. On a short nose the tangent profiles have to turn the flow harder over their whole length, and this trade-off favours the power-$\frac{1}{2}$ shape. Wildhorn chose this shape in 2022 on the basis of OpenRocket simulations.

**Integration and manufacture.** 217 mm of the 3U fits inside (nose bay 943 mm). The Firehorn process (prepreg on a PET-CF male mould, removable tip) exists and only needs scaling. The blunt apex naturally ends on a tip insert.

**Verdict: runner-up.** It is 22 m below Von Kármán at $L/D = 4.5$ (about 0.15 kg) and within 31 m of it in every robustness scenario (Section 4.5). It is the recommended fallback if reusing Firehorn tooling and know-how outweighs the shoulder kink and the 36 mm longer nose bay.

## 5.5 Power series $n = \frac{3}{4}$

**Geometry.** It lies between the cone and the $\frac{1}{2}$-power nose: a sharp-ish tip ($74^\circ$ apex tangent over a very short distance; the slope is infinite at the apex but the radius of curvature is zero, so the tip is pointed), a slender body, and the second-smallest volume (8.2 L) and wetted area. It has a $4.8^\circ$ shoulder kink.

**Flow.** Its wave drag falls fastest with Mach number ($0.0513 \rightarrow 0.0432$ between $M = 1.2$ and 1.6). Its slender, nearly conical body behaves like a cone whose shock weakens quickly as the Mach number rises, while its pointed tip avoids the stagnation spike of the power-$\frac{1}{2}$ nose. Crowell's chart rates it among the best around $M \approx 1.5$–1.9. Its low wetted area saves friction, so over the Weisshorn flight it ends 18 m below Von Kármán at $L/D = 4.5$.

**Integration.** It has the worst packaging of the smooth shapes: 152 mm of the 3U inside, a 1008 mm nose bay and a kinked shoulder.

**Verdict: not selected.** Its drag is competitive, but it would only make sense on a faster vehicle ($M_{\max} \approx 2$) with a payload outside the nose.

## 5.6 LV-Haack ($C = \frac{1}{3}$)

**Geometry.** The Haack nose that minimises wave drag for a given length *and volume*. It is the fullest design (11.6 L), with a tangent shoulder. It is the natural choice when the internal volume, rather than the base diameter, is fixed by the payload.

**Flow.** It is close to the ogive, with a stronger compression at the apex. Its wave drag is 13–19 % above Von Kármán over $M = 1.2$–1.6.

**Integration.** It has the best payload room: 320 mm of the 3U inside, a 870 mm nose bay, and the heaviest shell (376 g). It also has the most forward centre of pressure ($-0.08$ cal), the largest reduction of the static margin among the candidates.

**Verdict: the choice if payload volume were the driver.** It costs 69 m against Von Kármán for the same bay length as the ogive, so between the two volume-optimised options it is marginally better than the ogive.

## 5.7 Von Kármán ($C = 0$, recommended)

**Geometry.** The Haack nose that minimises wave drag for a given length *and base area*. Its apex has an infinite slope but a vanishing radius ($y \propto x^{3/4}$ near the tip), so in practice it is a slender needle that thickens quickly. Unlike the LV-Haack nose it does not constrain the volume, so it can stay thinner near the tip and spread the area growth more evenly along the length, which is what lowers its wave drag. It has 10.3 L of volume, a tangent shoulder and $x_{cp}/L = 0.5$.

**Flow.** The compression is weak and concentrated in the first few per cent of the length. The wall pressure then decreases monotonically and is below ambient over the aft third (Figure 6), which partly cancels the forward overpressure. It has the lowest wave drag of the seven at every computed Mach number at $L/D = 4.5$ (0.0437, 0.0425 and 0.0414 at $M = 1.2$, 1.4 and 1.6). The 4.5 mm spherical tip replaces the needle apex by a stagnation region a few millimetres across (as on any blunt tip), at a cost of $+0.5$ to $+1.4\,\%$ in wave drag (direct runs at $L/D = 4.5$).

**Integration and manufacture.** 254 mm of the 3U fits inside the curved part (nose bay 907 mm). The tangent shoulder gives a clean bond line for the Al coupler. The profile is generated from one closed-form equation (CSV/STL provided).

**Verdict: recommended.** It gives the highest apogee at $L/D = 4.5$ in all seven robustness scenarios, has no shoulder kink and gives reasonable payload room.

### 5.8 Literature cross-check

The drag-rating chart compiled by Crowell (reproduced on Wikipedia's "Nose cone design" page, CC BY-SA 4.0, and used in the Firehorn nosecone DJF) ranks the shapes from 1 (superior) to 4 (inferior) against Mach number. It condenses published drag comparisons into four qualitative classes and is the reference the Firehorn DJF used to choose its nose; it therefore provides an independent, if coarse, check of the computed ranking. Table 6 lists the ratings read from the chart at five Mach numbers. The chart does not state the fineness ratio of the data behind it.

Table 6: Crowell's qualitative drag ratings (1 superior, 2 good, 3 fair, 4 inferior; –: no data), read from the vector chart `Nose_cone_drag_comparison.svg` (G. A. Crowell / C. M. G. Lee). Compared with the ranking computed here at $L/D = 3$ and $L/D = 4.5$.

| | Crowell rating at $M =$ | | | | | Euler rank at $M = 1.4$ | |
|---|---|---|---|---|---|---|---|
| Shape | 0.9 | 1.0 | 1.2 | 1.5 | 1.85 | $L/D = 3$ | $L/D = 4.5$ |
| Cone | 4 | 4 | 4 | 4 | 4 | 6 | 7 |
| Tangent ogive | 4 | 4 | 4 | 4 | 4 | 5 | 6 |
| Parabola | 1 | 3 | – | – | – | 4 | 5 |
| Power $\frac{1}{2}$ | 1 | 2 | 2 | 2 | 3 | 1 | 2 |
| Power $\frac{3}{4}$ | 2 | – | – | 3 | 2 | – | 3 |
| LV-Haack | 1 | 1 | – | – | – | 3 | 4 |
| Von Kármán | 1 | 1 | 1 | 3 | 2 | 2 | 1 |

The two sources agree on what matters most: the ogive and the cone are the worst choices at transonic and low supersonic speeds, and the Von Kármán, Haack and power-law noses are the good ones. They differ around $M \approx 1.5$, where the chart prefers the $\frac{1}{2}$-power nose to Von Kármán. The Euler results reproduce that preference at $L/D = 3$ (power $\frac{1}{2}$ ahead) but not at $L/D \geq 4$, where Von Kármán is lower at every Mach number. The ranking depends on fineness, and the chart is a coarse, fineness-free summary. The recommendation therefore rests on the computed ranking at the actual design fineness, which is also robust to the modelling assumptions (Section 4.5).

## 6 Selection

### 6.1 Profile

Table 7 scores the profiles at $L/D = 4.5$ on the criteria of Section 2. Heating is left out because, with a metallic tip, it does not depend on the profile (Section 4.7).

The matrix is a weighted-sum decision method. Each design receives a score $s_i$ from 1 (worst) to 5 (best) on each criterion $i$, and its total is $\sum_i w_i s_i$ with weights $w_i$ that add up to one. The quantitative criteria are scored linearly between the worst and the best design: performance on the apogee change $\Delta z$, integration on the length of 3U inside the curved part, and stability on the change in static margin. Manufacture and heritage are judgement scores. For manufacture, a tangent shoulder and a standard mould score 5, a shoulder kink is penalised, and the cone gets 3 for its simple, developable mould. For heritage, the profile of the current Firehorn vehicle scores 5, the ogive of earlier vehicles 4, and profiles without ERT flight history lower. Performance carries the largest weight (0.4) because apogee margin is what the nose shape mainly affects and what systems engineering trades against mass growth. Integration and manufacture (0.2 each)

decide whether the design can be housed and built. Heritage and stability (0.1 each) matter less: heritage lowers risk without changing the hardware, and the static margin varies by less than 0.1 cal between candidates.

Table 7: Trade-off matrix at $L/D = 4.5$ (1 = worst, 5 = best). Performance: apogee at equal length (Table 3). Integration: 3U length inside the curved part. Manufacture: tangent shoulder, mould and tip. Heritage: flown or built at ERT. Stability: change in static margin.

| Criterion (weight) | Cone | Ogive | Parab. | Power $\frac{1}{2}$ | Power $\frac{3}{4}$ | LV-Haack | Von Kármán |
|---|---|---|---|---|---|---|---|
| Performance (0.40) | 1.0 | 2.6 | 2.6 | 4.3 | 4.4 | 2.9 | 5.0 |
| Integration, 3U fit (0.20) | 1.0 | 4.8 | 4.8 | 3.0 | 1.7 | 5.0 | 3.7 |
| Manufacture (0.20) | 3.0 | 5.0 | 5.0 | 3.0 | 2.0 | 5.0 | 5.0 |
| Heritage (0.10) | 1.0 | 4.0 | 2.0 | 5.0 | 1.0 | 1.0 | 2.0 |
| Stability (0.10) | 5.0 | 1.4 | 1.5 | 2.2 | 3.9 | 1.0 | 2.1 |
| **Weighted total** | 1.80 | 3.54 | 3.36 | 3.64 | 3.00 | 3.37 | **4.15** |

**Von Kármán is selected.** Three profiles form a leading group on flight performance: Von Kármán, power $\frac{1}{2}$ and power $\frac{3}{4}$. At the design length, Von Kármán is first. It has the lowest wave drag at every computed Mach number, flies 18 m higher than power $\frac{3}{4}$ and 22 m higher than power $\frac{1}{2}$, and stays first in all seven robustness scenarios (Figure 10). Ogive, parabolic and LV-Haack lose 69 m to 79 m, and the cone 132 m. Von Kármán also wins on the other criteria:

- it is tangent to the body at the shoulder, so there is no kink at the coupler bond. The power-$\frac{1}{2}$ nose has a $3.2°$ kink and the power-$\frac{3}{4}$ a $4.8°$ kink at $L/D = 4.5$;
- it holds more of the 3U unit (254 mm inside the curved part, against 217 mm for power $\frac{1}{2}$ and 152 mm for power $\frac{3}{4}$);
- its advantage holds at $M \geq 1.2$, so it does not hinge on the transonic model.

The Firehorn power-$\frac{1}{2}$ nose remains an acceptable alternative (22 m lower) if reusing its tooling outweighs the shoulder kink (R10, Section 7).

The outcome depends on the weights in one case only. If the weights of performance and integration were swapped (payload room 0.4, performance 0.2), the tangent ogive would score higher than Von Kármán (3.98 against 3.89). The selection therefore rests on the premise that apogee margin matters more than payload room, which holds as long as the payload is a single 3U unit that the Von Kármán nose can house. If the payload grows, the volume-driven alternatives of Table 9 apply.

## 6.2 Length

Every 90 mm of nose ($\Delta L/D = 0.5$) around $L/D = 4$–4.5 is worth 60 m of apogee, about 0.4 kg of mass margin at –143 m/kg. It also costs about 60 mm of nose-bay length (the extra length is partly recovered because more of the 3U fits inside the curved part) and about 30 g of shell. With the LV capped at 5.4 m (2027_C_SE_REQ_04) and dry-mass growth expected (the LVS flags its own mass as underestimated; 2027_C_SE_REQ_06 caps it at 50 kg), the design point is $L/D = 4.5$**,** $L =$ **810 mm**, which gives 60 m of margin for 62 mm of bay length. If systems engineering cannot release those 62 mm, the same profile at $L/D = 4$ (720 mm, the LVS length) is the fallback; nothing else in the design changes.

The trade-off behind this choice is between apogee and vehicle length. The wave drag falls as $1/(L/D)^2$, so each extra 90 mm is worth less than the previous one, while the length, wetted area and mass it adds grow linearly. A further step to $L/D = 5$ would still add about 39 m (Table 5) for about 60 mm more bay. $L/D = 4.5$ is the step requested here because every millimetre comes out of a 5.4 m budget that is already at its cap. The same reasoning can be extended if systems engineering finds more length.

### 6.3 Tip

A spherical tip of radius $r_n = 4.5$ mm ($r_n/R = 0.05$), machined in Al 6082-T6 as a replaceable insert:

- it removes the sharp apex that cannot be laid up in prepreg and chips on landing;
- it costs $+0.6\,\%$ in wave drag (–2 m of apogee). A 9 mm tip would lower the stagnation heating by $\sqrt{2}$ but cost –10 m, and the aluminium tip does not need it;
- it stays below 73 °C in the envelope case, whereas a printed PET-CF tip reaches 138 °C;
- it adds about 60 g at the very front, which helps the static margin.

The radius is a compromise between drag and heating. At $r_n/R = 0.05$ the drag cost is about 1 % or less over the Mach range, whereas $r_n/R = 0.1$ already costs about 4 % (Figure 11a). A smaller radius would raise the stagnation heating ($h_s \propto r_n^{-1/2}$) and make the tip more fragile to machine and handle. Because the aluminium insert keeps the tip cool at this radius, nothing is gained by blunting it further.

## 7 Recommendations

### 7.1 Recommended design

**Von Kármán nose cone, $L = 810$ mm ($L/D = 4.5$), $D = 180$ mm, tangent 60 mm shoulder, 4.5 mm spherical Al 6082-T6 tip insert, CFRP $[0/45/0]_{SO}$ shell (1.2 mm).**

Predicted: $C_{D,\mathrm{w}} = 0.0425$ at $M = 1.4$ (lowest of seven profiles at every computed Mach number); 60 m of apogee against the LVS-length baseline; 354 g for shell and tip; 254 mm of the 3U inside the curved part; tip below 73 °C and shell inner face below 104 °C on a 30 °C day.

## 7.2 Numbered recommendations

Table 8: Recommendations, owners and timing.

| # | Recommendation | Owner | When |
|---|---|---|---|
| R1 | Adopt the Von Kármán profile (Haack $C = 0$). Use `nose_profile_final.csv` or the STL in `results/cad/` as the master geometry. | FD $\rightarrow$ ST | PDR |
| R2 | Book a 907 mm nose bay (810 mm curved + 97 mm cylinder for the 3U, of which 60 mm is the coupler shoulder) in the 5.4 m LV budget. If it does not fit, keep the same profile at $L = 720$ mm ($-60$ m, bay 845 mm). | SE | PDR |
| R3 | Use a machined Al 6082-T6 tip insert ($r_n = 4.5$ mm, $\sim$80 mm long, $\sim$60 g), threaded onto a ring bonded inside the shell. Do not fly a printed PET-CF tip: it would exceed its heat-deflection range by 25 °C to 70 °C. | ST | CDR |
| R4 | Lay up the shell in CFRP prepreg $[0/45/0]_{SO}$, 1.2 mm, on a male PET-CF mould printed in 2–3 segments, with the joints in the aft half (low curvature). Use GFRP instead if the nose has to be the RF window. | ST (AV, PL) | CDR |
| R5 | Raise the thermal requirement to 430 K for 10 s and qualify the shoulder bond: the DP490 joint at the shoulder sees about 95 °C in the envelope case. Consider a higher-$T_g$ adhesive if the test margin is small. | SE, ST | CDR |
| R6 | Verify with Fluent (RANS, k-$\omega$ SST) at $M = 0.95$, 1.2 and 1.47 for the selected nose and the power-$\frac{1}{2}$ alternative, plus one $5^\circ$ incidence case for $C_{N\alpha}$ and CP. This replaces the transonic model of this report. | FD-CFD | before CDR |
| R7 | Put the final geometry and $C_{D,\mathrm{w}}(M)$ into the RocketPy model. Re-check apogee, rail-exit speed and the 2–4 cal static margin once the fins are designed. The nose contributes $C_{N\alpha} = 2$ at 405 mm from the tip. | FD-Sim | PDR |
| R8 | Size the payload structure around a 3U envelope whose front face is 254 mm ahead of the shoulder, inside the $r \leq 77$ mm contour (Figure 15). | PL, ST | CDR |
| R9 | Create the missing Weisshorn nosecone requirements: geometry, bay length, mass ($\leq 500$ g for the nose structure, Table 11), thermal and tip (Section 9). | SE | PDR |
| R10 | Keep the Firehorn power-$\frac{1}{2}$ profile as the documented alternative. It costs 22 m of apogee and adds a $3.2^\circ$ shoulder kink, but its tooling and process already exist. | FD | – |

Owners are the ERT teams (Nomenclature); PDR and CDR are the preliminary and critical design reviews. R6 asks for a viscous check with RANS and the $k$-$\omega$ SST turbulence model, a two-equation model widely used for external aerodynamics with adverse pressure gradients and shock–boundary-layer interaction. It covers the transonic range, where the Euler model of this report does not apply (Section 3.2), and gives the normal force at incidence, which Barrowman's method only estimates. R7 transfers the results to RocketPy, the team's six-degree-of-freedom trajectory simulator, which also models the rotation of the vehicle.

## 7.3 When to choose another design

The recommendation assumes the current Weisshorn priorities: apogee margin, a 3U payload in the nose bay, and a length cap. If those priorities change, Table 9 gives the design to use instead.

Table 9: Choice of nose cone if the priorities change.

| If . . . | choose | cost relative to the recommendation |
|---|---|---|
| the LV cannot grow beyond the LVS length | Von Kármán, 720 mm | $-60$ m apogee |
| schedule favours reusing Firehorn tooling | Power $\frac{1}{2}$, 810 mm | $-22$ m, $3.2°$ kink, bay $+36$ mm |
| the payload grows (more volume in the nose) | LV-Haack, 810 mm | $-69$ m, bay $-37$ mm, $+22$ g |
| the nose must be as short as possible ($L/D \leq 3.5$) | Power $\frac{1}{2}$ | lowest drag at short length |
| the vehicle becomes faster ($M_{\max} \approx 2$) and the payload leaves the nose | Power $\frac{3}{4}$ or Von Kármán at $L/D \geq 5$ | re-run the study |
| a subsonic test vehicle (e.g. Aletsch) | Tangent ogive or Von Kármán | shape barely matters below $M = 0.8$ |

Each row follows from the results above. A shorter vehicle keeps the same profile and gives up the length gain of Section 6. The power-$\frac{1}{2}$ nose is the closest alternative in performance and the only one with an existing process. If the payload volume becomes the driver, the LV-Haack nose, the minimum-drag shape for a given volume, houses the most payload for the smallest drag penalty. On short noses the power-$\frac{1}{2}$ shape has the lowest wave drag (Section 4.3). At higher Mach numbers the wave drag of the power-$\frac{3}{4}$ nose falls fastest (Section 5), so the ranking has to be recomputed. Below $M = 0.8$ there is no wave drag, and the choice is made on volume and manufacturing alone.

## 7.4 What not to do

- **Do not use a cone or a tangent ogive** for a supersonic Weisshorn. They lose 79 m to 132 m at the same length, and both this study and the literature rate them worst in the transonic and low supersonic range.
- **Do not use a printed thermoplastic tip.** Its surface follows the stagnation temperature, which reaches 156 °C on a hot day.
- **Do not size the nose by fineness alone.** A longer nose pays only if the extra length also houses payload. Measure it as nose-bay length, which is what the 5.4 m cap actually constrains.

# 8 Design Definition

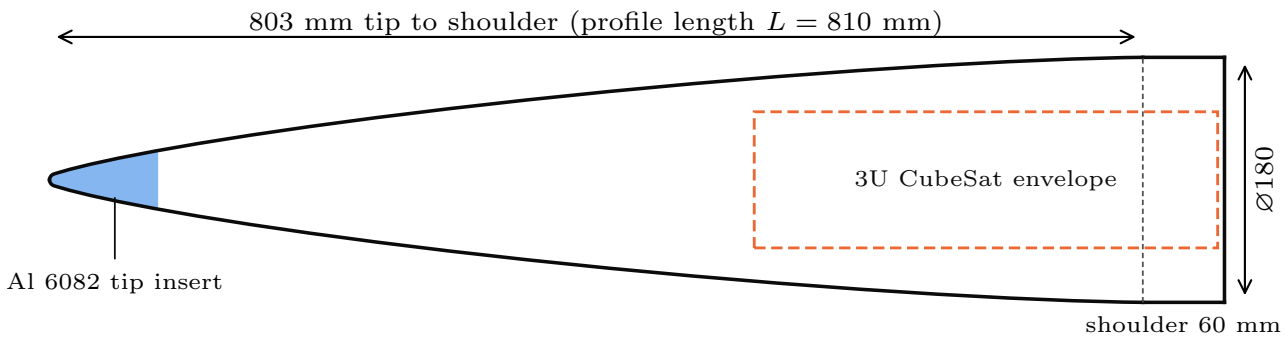



Figure 15: Selected nose cone to scale: Von Kármán profile, $L/D = 4.5$, 4.5 mm spherical Al 6082 tip, 60 mm bonded shoulder; dashed: 3U CubeSat envelope.

Table 10: Nose cone definition (outer mould line and performance).

| | |
|---|---|
| Profile | Von Kármán (Haack series, $C = 0$): $y = \frac{R}{\sqrt{\pi}}\sqrt{\theta - \frac{1}{2}\sin 2\theta}$, $\theta = \arccos\left(1 - \frac{2x}{L}\right)$ |
| Base diameter $D = 2R$ | 180 mm |
| Profile length $L$ (to the theoretical apex) | 810 mm ($L/D = 4.5$) |
| Tip | sphere $r_n = 4.5$ mm, tangent at $x = 9.8$ mm; apex truncated by 6.7 mm |
| Physical curved length (tip to shoulder) | 803.3 mm |
| Shoulder | 60 mm cylinder, Al male coupler bonded inside (Firehorn design) |
| Angle to the body at the shoulder | $0°$ (tangent) |
| Internal volume / wetted area | 10.31 L / 0.300 m$^2$ |
| Shell | CFRP prepreg $[0/45/0]_{SO}$, 1.2 mm, on a PET-CF male mould (Firehorn process); GFRP if the nose must be an RF window |
| Tip insert | Al 6082-T6, machined, threaded onto a bonded Al ring (Matterhorn/Firehorn concept), $\sim 60$ g, replaceable |
| Estimated mass (shell + tip, without coupler) | 354 g |
| Nose CP (Barrowman, from the apex) | 405 mm, $C_{N\alpha} = 2$ |
| 3U inside the curved part | 254 mm; the remaining 97 mm of the 3U sits behind the shoulder |
| Nose wave drag $C_{D,\mathrm{w}}$ at $M = 1.2/1.4/1.6$ | 0.0437 / 0.0425 / 0.0414 (Euler at $L/D = 4.5$, sharp apex; $+0.5$ to $+1.4$ % with the tip) |
| Apogee (nominal, model) | 9060 m, i.e. 60 m vs. the LVS-length baseline; $M_{\max} = 1.47$ |
| Static margin change vs. baseline | -0.04 cal |
| Peak temperatures (envelope) | tip 73 °C; shell inner face 104 °C |

The profile length $L$ is the length of the mathematical profile up to its theoretical apex. The 4.5 mm sphere is tangent to the profile at $x = 9.8$ mm from that apex. Replacing the apex by the sphere removes 6.7 mm of length, hence the 803.3 mm physical curved length. The coordinates are in `results/nose_profile_final.csv` (200 points, $x$ from the tip, clustered at the tip). Two notes for Structure:

- **Mould length.** The profile is longer than one print bed, so the PET-CF male mould needs 2–3 segments. Joints should sit where the curvature is low, i.e. in the aft half.
- **Tolerance.** $\pm 0.5$ mm on the radius moves $C_{D,\mathrm{w}}$ by much less than 1 %. A step at the tip/shell joint matters more: keep it below 0.2 mm and forward-facing steps at zero. A forward-facing step acts as a small blunt face in the supersonic flow: it creates a local shock with a local pressure and heating peak, and it can trip the boundary layer.

## 8.1 Manufacturing plan

1. **Mould.** Male mould printed in PET-CF from the STL (Firehorn process). The 870 mm part (curved length + shoulder) is printed in three segments, joined at $x \approx 330$ and 600 mm, where the radius changes slowly. The segments are aligned on a central rod, sanded, and sealed with release agent.
2. **Layup.** Prepreg $[0/45/0]_{SO}$ (1.2 mm), the Firehorn laminate. Plies are cut from flat patterns and stop at $x \approx 60$ mm, where the radius is already about 17 mm. This avoids the ply pile-up at the apex reported on Firehorn: as the circumference shrinks towards the apex, the fabric can no longer lie flat and the plies overlap. Two extra local plies are added over the last 100 mm of the curved part and the shoulder, where the coupler is bonded and the shell is hottest at the bond.
3. **Tip.** Machined Al 6082-T6 insert covering the first 80 mm, threaded onto an Al ring that is bonded inside the shell at the step $x = 60$ mm (Matterhorn/Firehorn concept). The outer surface is flush with the shell within 0.2 mm.
4. **Coupler.** Firehorn male glued coupler, scaled to 180 mm, bonded on the 60 mm shoulder (2027_C_ST_COUPLERS).
5. **Finish.** Sand and paint. Check the profile with a printed template (or a CMM) at five stations against `nose_profile_final.csv`, tolerance $\pm 0.5$ mm.

## 8.2 Mass budget

Table 11: Mass budget of the nose cone structure (without payload, payload bay and coupler).

| Item | Mass [g] | Basis |
|---|---|---|
| CFRP shell, 0.300 $m^2$ at 0.98 $kg/m^2$ | 294 | Firehorn as-built areal density |
| Local reinforcement plies (last 160 mm) | 35 | 2 plies over 0.09 $m^2$ |
| Al 6082-T6 tip insert | 60 | machined, hollowed |
| Bonded Al tip ring + fastener | 20 | estimate |
| Adhesive (tip ring, coupler) | 25 | estimate |
| Paint and filler | 15 | 0.05 $kg/m^2$ |
| **Total** | **449** | |
| Proposed requirement | 500 | 11 % margin |

## 8.3 CAD deliverables

The folder `results/cad/` contains watertight binary STL solids (units mm, $x$ from the tip, closed by a flat base at the end of the shoulder) for the seven candidates at $L/D = 4.5$ and for the final design with its 4.5 mm tip (`nose_vonkarman_LD4.5_rn4.5mm_FINAL.stl`). They are checked to be closed manifolds whose volume matches the analytical volume to 0.05 %. The same files can be imported directly into SpaceClaim for the Fluent runs of R6.

# 9 Open Points and Next Steps

1. **Fluent verification (CFD team).** Run the selected geometry in the team's Fluent workflow (density-based, k-$\omega$ SST, ideal gas, Sutherland, far field 10–20$D$ upstream and radially, 30–50$D$ downstream) at $M = 0.95$, 1.2 and 1.47. This replaces the transonic assumption of Section 3.2 and checks the Euler wave drag with viscous effects. A density-based solver solves the compressible equations in coupled form and is the usual choice for flows with shocks. Sutherland's law gives the viscosity of air as a function of temperature. The far-field distances keep the boundaries clear of the bow shock and the wake, so that the imposed free-stream conditions do not disturb the solution. The Euler wall-pressure files (`results/cp/`) are the reference to compare against.

2. **RocketPy update (simulation team).** Replace the LVS nose in the RocketPy model by the final geometry and its $C_D(M)$ (`results/euler_results.csv`), then re-check apogee, static margin (2–4 cal, 2027_C_SE_FD_REQ_07) and the 30 m/s rail exit speed once the fins are defined.

3. **Requirements to create** (none exists for the Weisshorn nosecone). Proposed:

   - profile: Von Kármán, $L = 810$ mm ($L/D = 4.5$, 803.3 mm physical with the tip), coordinates from `nose_profile_final.csv`, radius tolerance $\pm 0.5$ mm;
   - nosecone bay length (curved part + the cylinder that completes the 3U housing) 907 mm, to be booked in the 5.4 m LV budget (2027_C_SE_REQ_04);
   - thermal: withstand 430 K on the outer surface for 10 s (envelope stagnation temperature, an upper bound for the whole surface), replacing Firehorn's 390 K/5 s;

- tip: metallic (Al 6082-T6), spherical radius 4.5 mm, replaceable.

4. **Material and RF.** If the payload or avionics need GNSS/telemetry through the nose, the shell must be GFRP (2027_C_SE_AV_INTERFACE_ST_REQ_07 asks for a $360°$ RF window $\geq 2D$ long). The nose is 810 mm long, so it can be that window. GFRP has the same epoxy temperature limit but about 40 % more mass for the same stiffness; the shape result does not change. Glass fibre is an electrical insulator and lets radio waves through, whereas the conductive carbon fibres screen them.
5. **Thermal test.** Update 2025_C_ST_NOSECONE_TEMPERATURE_TSP to the new level and add a shoulder/coupler bond specimen. The DP490 bond is the weakest point in the envelope case (Section 4.7).

# A Scripts and data

All files are in the working folder `Nose Cone/` of this study. Python 3.14, NumPy, SciPy, pandas, Matplotlib; no other dependency. Table 12 lists the scripts that implement the method chain of Section 3 and the result files they produce.

Table 12: Scripts and result files.

| File | Content |
|---|---|
| `scripts/geometry.py` | profile equations, spherical blunting, volume, wetted area, CP, 3U fit |
| `scripts/euler_axi.py` | axisymmetric Euler solver (grid generator, HLLC/MUSCL, BCs, drag integration) |
| `scripts/taylor_maccoll.py` | exact conical flow, used for validation |
| `scripts/run_sweep.py` | parallel sweeps: `main` (7 shapes $\times$ 4 fineness $\times$ 5 Mach), `cones`, `grid`, `blunt` |
| `scripts/flight.py` | atmosphere, component drag build-up, point-mass ascent |
| `scripts/thermal.py` | stagnation and turbulent heating, 1D wall conduction, lumped metal tip |
| `scripts/analysis.py` | calibration of $k$, trade table (apogee, mass, CP shift, 3U fit) |
| `scripts/figures.py`, `figstyle.py` | all figures of this report |
| `results/euler_results.csv` | every converged Euler point: shape, $L/D$, $r_n/R$, grid, Mach, $C_{D,\mathrm{w}}$ (rebuilt from `cp/`) |
| `scripts/flowfield.py` | direct Euler runs at $L/D = 4.5$ with flow-field output |
| `scripts/robustness.py` | ranking under perturbed modelling assumptions |
| `scripts/cad.py` | STL solids of all designs |
| `scripts/figures_extra.py` | design gallery, pressure fields, drag breakdown, robustness |
| `results/fields/*.npz` | flow fields at $L/D = 4.5$, $M = 1.4$ and 1.2 |
| `results/robustness.csv` | apogee of every shape in every scenario |
| `results/cad/*.stl` | watertight STL solids (mm) of the seven designs and the final nose |
| `scripts/make_tables.py` | tables and every number quoted in the text (`sections/macros.tex`) |
| `results/euler_sweep.csv` | run log of completed jobs with iteration counts |
| `results/cp/*.npz` | wall pressure distributions of every run |
| `results/trade_table.csv` | trade-study table (Section 4.3) |
| `results/nose_profile_final.csv` | final profile coordinates for CAD |

Reproduce everything with:

```
python run_sweep.py main 6 ; python run_sweep.py cones 3
python run_sweep.py fast 7 ; python run_sweep.py fast2 3 ; python flowfield.py
python analysis.py ; python robustness.py ; python make_tables.py
python figures.py ; python figures_extra.py ; python cad.py
```